

Common Mass-Charge Matching and Universal Inertial–Active Proportionality under AASC

A Poincaré–Noether Form-Factor-to-Bargmann Overlap Theorem from a
Normalized-Ray Module A Source Lift, Polarization-Attached Total EMT,
Exact Zero-Transfer Ward Normalization, Complete Composite Closure, and
an Explicit Wigner Contraction

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Module identity. AASC-GRQM-MB-MASS-1.0, v1.3.0-normalized-ray-source-lift-corrected.

Program status. Standalone normalized-ray-source-lift-corrected Module B of the frozen modular repair program for *Complete Certificate Closure for a Constraint-Native Gravity–Quantum Realization under AASC*. Accepted Module A is an immutable theorem-use input. The present correction repairs the projective-state and source-representative types of the operator/source bridge. Physical source evaluation is now a family on normalized rays formed at the Module A prequotient level; one fixed concrete representative family is produced before lawful quotienting. Its norm-squared homogeneous vector extension, not the physical ray map itself, is polarized to reconstruct the unique operator energy–momentum distribution. The form-factor and charge branches are required to use that operator.

Abstract

This manuscript proves the mass-charge bridge required by the constraint-native AASC gravity–quantum realization program. The dependency order is fixed. Target adequacy first forces the AASC kernel; the realized physical interior, particle persistence, deformation-response mass role, gravity role, and quantum-standing role are fixed next; only afterward may Poincaré, Einstein, Schrödinger, and accepted metric-source representations enter. No energy–momentum form factor, Poincaré Casimir, active metric monopole, Bargmann central charge, or numerical mass coordinate is permitted to authorize the upstream role it represents.

The exact domain is the operator-presentable refinement

$$\widehat{Q}_{\text{mass}}^{\text{op}} \longrightarrow \widehat{Q}_{\text{mass}} = F^{-1}(Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}}) \subseteq \widehat{Q}_{\text{src}},$$

where the first map forgets a common quantum domain, a total normalized-ray family of accepted Module A source presentations, its coherent concrete prequotient representative family, the norm-squared homogeneous vector extension, and the unique operator-valued distribution reconstructed from that extension. This refinement is then equipped with the frozen scalar Poincaré–Noether/form-factor/Breit/Bargmann certificate. The certificate contains a full irreducible vacuum/reference representation, a conserved total energy–momentum operator whose global charges implement translations, a stable positive-mass spin-zero spectral sector, a complete total-EMT form-factor family, a regulated stationary Breit source profile, a linearized Einstein first-response envelope, and a standard scalar Wigner representation with an explicit common-core nonrelativistic contraction.

The first new theorem is the normalized-ray operator/source attachment theorem. A physical source family is defined on $\mathbb{P}_1(\mathfrak{D}_x)$, the unit sphere modulo phase. Each ray is sent by one fixed Module A state-evaluation protocol to a concrete prequotient source presentation, whose unique distributional representative is constructed before lawful source quotienting. For an arbitrary vector ψ , the auxiliary extension

$$\mathbf{T}_{x,\psi}^{\sharp} = \|\psi\|^2 \mathbf{T}_{x, [\psi/\|\psi\|]}^{\text{mray}}$$

is homogeneous of degree two but is not itself a physical-state map. When its diagonal pairings satisfy the parallelogram and continuity certificate, complex polarization reconstructs a unique Hermitian operator-valued distribution

$$\widehat{\mathbf{T}}_x : \Gamma_c(S^2 T^* M) \longrightarrow \mathcal{L}(\mathfrak{D}_x, \mathfrak{D}'_x)$$

whose normalized diagonal quadratic forms are exactly the concrete accepted Module A ray-state sources. Support, naturality, Ward, boundary, and improvement ledgers lift by homogeneous extension and polarization. An unrelated operator current or state-dependent choice of source representative cannot satisfy the attachment relation.

The next theorem is global. The integrated charge of the attached total energy–momentum tensor and the Stone translation representation are compared on the full vacuum reference representation. They implement the same translation automorphisms, so their unitary groups differ only by a commutant phase. Irreducibility and the invariant baseline remove that phase. Only then is the equality restricted to the stable mass sector; the local observable algebra and the vacuum are never placed inside a one-particle subspace.

The second new theorem uses the total energy–momentum form factor rather than a selected-state expectation four-momentum. For covariantly normalized generalized one-particle states in a stable spin-zero sector, the matrix element is taken from that same attached operator:

$$\langle p' | \widehat{\mathbf{T}}_x^{\mu\nu}(0) | p \rangle = 2\bar{P}^\mu \bar{P}^\nu A(t) + \frac{1}{2}(\Delta^\mu \Delta^\nu - \eta^{\mu\nu} \Delta^2) D(t).$$

The global charge identity forces the exact Ward normalization $A(0) = 1$. A regulated Breit-frame distribution constructed from this complete total tensor is stationary and conserved, has integrated

energy $m_{\text{Cas}}c^2$, and satisfies exact Laue closure $\int \mathcal{T}_x^{ij} d^3x = 0$. The linearized Einstein solution therefore has

$$h_{00}(r) = \frac{\kappa c^2 m_{\text{Cas}}}{4\pi r} + O(r^{-2}),$$

while the internal-stress form factor contributes no independent $1/r$ monopole. The result applies to the total tensor of a stable composite, so binding, mediator, interaction, internal-stress, restoration, transport, baseline, and boundary terms are included through accepted Module A rather than replaced by a sum of bare constituent masses.

The third new theorem constructs the inertial branch explicitly. The massive scalar Wigner representation is realized on $L^2(\mathbb{R}^3, mc^2 E_c(\mathbf{p})^{-1} d^3p)$, unitarily transported to $L^2(\mathbb{R}^3, d^3p)$, and restricted to a common smooth core. The rest-subtracted Hamiltonian and boost generators converge to

$$H_{\text{nr}} = \frac{\mathbf{P}^2}{2m_{\text{Cas}}}, \quad G_i = i\hbar m_{\text{Cas}} \partial_{p_i},$$

with commutator convergence $[G_i, P_j] = i\hbar \delta_{ij} m_{\text{Cas}} I$. Thus the Bargmann central mass and the minimal scalar Schrödinger kinetic denominator equal the same Poincaré Casimir mass.

Consequently the active soft-gravitational and inertial Hamiltonian readouts factor through one common Casimir charge line and satisfy

$$m_g(x) = \alpha_D m_i(x)$$

with one projection-wide factor fixed solely by the common metric and inertial calibrations. The translation Ward normalization $A(0) = 1$ excludes an object-dependent gravitational monopole factor inside the frozen total-EMT scope. The common Einstein–Newton/Bargmann calibration gives $m_g = m_i$. Naturality and flat calibration transport preserve the relation on the declared calibration-connected envelope.

The manuscript does not identify the Casimir with the invariant norm of an arbitrary state’s mean four-momentum, does not require a normal state to be simultaneously sharp in momentum and spatially localized, does not assert a conserved global energy on generic curved spacetime, and does not prove strong equivalence, self-force, all-order self-gravity, numerical mass values, Module C continuation, or Module D orbit closure.

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$$\text{TargetAdeq}_D \implies \mathbf{K}_D \implies A_D^+ \implies \mathbf{l}_D \implies (Q_{\text{resp}}, Q_{\text{in}}).$$

Accepted Module A then supplies

$$Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}, \quad q_g : \widehat{Q}_{\text{src}} \longrightarrow \text{Src}_g^{\text{dist}} / \simeq_{\text{law}}.$$

Only on

$$\widehat{Q}_{\text{mass}}^{\text{op}} \longrightarrow \widehat{Q}_{\text{mass}} = F^{-1}(Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}})$$

may Module B add the exact stable-sector certificate. The corrected proof order is

$$\begin{aligned} &\text{operator/source attachment} \implies \text{attached total-EMT charge} \\ &\implies \text{global translation-generator equality} \implies m_{\text{Cas}}. \end{aligned}$$

$$m_{\text{Cas}} \implies \begin{cases} A(0) = 1 \implies \text{Breit source} \implies m_g, \\ \text{explicit Wigner contraction} \implies M_B = m_{\text{Cas}} I \implies m_i, \end{cases} \implies m_g = \alpha_D m_i.$$

The old inference $-\omega(P)^2 = \omega(-P^2)$ is neither used nor permitted.

Exact claim status

Item	Status in this manuscript
Kernel, RPI, mass/inertia, gravity, quantum standing	Imported theorem-use packages; never authorized by a mass coordinate.
Accepted metric-source representation and finite composite closure	Imported from accepted Module A, hash locked.
Operator/source attachment	New theorem from normalized-ray Module A source presentations, a coherent prequotient representative family, norm-squared homogeneous extension, and complex polarization.
Global attached-source-charge/translation-generator equality	Proved on the full irreducible reference representation, then restricted spectrally.
Stable total-EMT form factor and $A(0) = 1$	New theorem proved from the global charge identity.
Stationary Breit profile, Laue closure, active monopole	New distributional theorems for the complete total EMT.
Scalar Wigner-to-Bargmann contraction	New explicit common-core construction and inhabitant.
Universal proportionality and calibrated equality	New theorems in the frozen matching and transport envelopes.
Generic curved energy, strong equivalence, self-force, unique orbit, upstream continuation	Explicit nonclaims.

Keywords

AASC; energy–momentum form factors; Poincaré mass; Ward normalization; Breit source; active gravitational mass; Bargmann central extension; Wigner representation; nonrelativistic contraction; Schrödinger mass; equivalence principle.

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Part I

Dependency Lock and Exact Matching Target

1 Proof class, fixed target, and dependency inversion

1.1 The target is a matching theorem, not a new ontology

The present module begins after the kernel, the realized physical interior, and the two downstream projection roles have already been fixed. Its target is narrower than a theory of all mass and stronger than a comparison of two symbols called “mass.” It asks whether, on the exact domain where both a certified metric-source representation and a stable inertial representation exist, the intrinsic scalar active metric response and the inertial scalar are two representations of one Poincaré–Noether charge.

Definition 1.1 (Module B target). The Module B target is the class of source-attached mass-bearing objects that admit all of the following without changing the upstream physical-interior target:

1. an upstream quotient-stable deformation-response class;
2. a transport-facing inertial witness;
3. an attachment-certified Module A metric-source witness;
4. a full positive-energy Poincaré representation with invariant baseline and an isolated stable positive-mass scalar sector;
5. an admitted total stress-energy current whose integrated charge generates translations on the full representation;
6. a regular elastic zero-energy-transfer stress-tensor form-factor channel on the stable sector;
7. the minimal Einstein first-response coupling of that complete total source;
8. an explicit common-core Poincaré-to-Bargmann contraction;
9. a common calibration line and lawful calibration transport.

The target also retains a separately typed stationary localized source-profile subclass for the Laue and far-field theorems. It does not require a normal state to be simultaneously a sharp momentum eigenstate and a spatially localized stationary source. Universal proportionality and calibrated equality are conclusions, not target clauses.

Theorem 1.2 (Dependency inversion for common mass charge). *Every nondegenerate use of inertial mass, active metric-source mass, Poincaré mass, Bargmann mass, or a zero-transfer gravitational form factor as a determinate same-domain comparison object already presupposes the kernel and*

the upstream response identity to which the quantity is attached. Therefore no mass coordinate or form factor used below can authorize the target, the kernel, the realized physical interior, or the deformation-response role.

Proof. Each quantity is used comparatively: it identifies the same source or inertial response across redescription, composition, transport, and calibration. Such use requires a fixed target (Reference), licensed reuse (Standing), a boundary between admissible and inadmissible comparison (Admissibility), and preservation of failed comparison as failed rather than retroactively repaired (Irreversibility). These are the kernel roles. The numerical or distributional output is therefore downstream from the object and witness family whose comparison it records. Using it to create that object would reverse the same dependency it presupposes. $\square$

1.2 Frozen module endpoint

The frozen standalone endpoint of this module remains:

1. construction of one common Poincaré–Noether charge on the matching subtype;
2. derivation of the active first-response metric monopole from the complete total source;
3. derivation of the Bargmann/Schrödinger inertial coefficient from the same intrinsic charge;
4. one projection-wide proportionality factor on the calibration-connected fixed envelope;
5. calibrated equality after a common source/inertial normalization;
6. explicit failure classification outside the matching hypotheses.

The Cycle-1 repair changes only the mathematical bridge used to discharge the endpoint. The invalid substitution of a selected state’s expectation four-momentum for the operator Casimir is removed. The common source charge is now obtained from exact zero-transfer normalization of the complete total stress-tensor form factor.

1.3 Correctness-relevant criticism

A criticism has standing only if it identifies a false premise, invalid inference, hidden import, scope shift, overreported conclusion, or broken named link in one of the following places:

- the mass-bearing type lock;
- the accepted Module A source import;
- the full-representation Poincaré action/generator/current/charge bridge;
- charge integrability, baseline fixation, or improvement invariance;
- stable-sector Casimir classification;
- generalized-state and zero-transfer form-factor normalization;
- complete-composite source typing and Laue closure;
- the linearized Einstein zero-transfer or stationary far-field calculation;

- the explicit common-core Wigner contraction;
- the common charge-line and calibration argument;
- naturality or transport extension;
- the final scope report.

Preference for a metric-first, Hilbert-first, or primitive equivalence-principle proof is not a correctness criterion unless it breaks one of these links.

2 Exact response, source, inertial, and mass-bearing types

The repair discipline begins by distinguishing the domains on which the two readouts are actually well typed and by separating intrinsic sector data from state- and profile-dependent quantities.

Definition 2.1 (Complete response quotient). Let Q_{resp} be the quotient of admitted deformation-response presentations by the complete standing-bearing response signature. Equality in Q_{resp} preserves every independently admitted deformation, transport, restoration, composition, scale-transport, and failure witness.

Definition 2.2 (Inertial subtype). The inertial subtype $Q_{\text{in}} \subseteq Q_{\text{resp}}$ consists of those response classes whose transport-facing restriction possesses a complete inertial witness in the declared projective-Hilbert or Poincaré sector. The subtype includes no claim that the inertial coordinate is already numerical.

Definition 2.3 (Accepted source-bearing subtype). The base source-bearing subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$ and its structured refinement $\hat{Q}_{\text{src}}$ are the objects constructed by accepted Module A. An element of $\hat{Q}_{\text{src}}$ retains an upstream response presentation, a metric-incidence attachment certificate, and its complete operational metric-response witness. The forgetful map is

$$F : \hat{Q}_{\text{src}} \rightarrow Q_{\text{resp}}^{\text{src}}.$$

Definition 2.4 (Mass-bearing overlap). Define

$$Q_{\text{mass}} := Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}}, \quad \hat{Q}_{\text{mass}} := F^{-1}(Q_{\text{mass}}) \subseteq \hat{Q}_{\text{src}}.$$

The first set is the exact upstream overlap on which both inertial and metric-source standing are available. The second retains the attachment and source-witness data required by the metric side.

Proposition 2.5 (No automatic totalization). *Neither inclusion*

$$Q_{\text{in}} \subseteq Q_{\text{resp}}^{\text{src}}, \quad Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{in}}$$

is asserted. A response may be source-bearing without a stable inertial particle representation, or inertial without entering the accepted metric-source scope.

Proof. The two subtypes are formed by different complete witness families. Module A admits a source only after a complete metric-incidence attachment exists. The inertial subtype requires a complete transport-sector witness. Neither witness family logically entails the other. The present module therefore works on their intersection and does not promote an absent witness by notation. $\square$

Definition 2.6 (Three noninterchangeable mass-related objects). For a stable sector and an admitted state/profile, keep the following objects distinct:

1. the intrinsic Casimir mass $m_{\text{Cas}} > 0$, defined by

$$-\eta_{\mu\nu} \hat{P}^\mu \hat{P}^\nu = m_{\text{Cas}}^2 c^2 I$$

on the stable sector;

2. the selected-state expectation four-momentum

$$p_\omega^\mu := \omega(\hat{P}^\mu);$$

3. the energy mass of a localized stationary source profile

$$M_{\text{src}} := E_{\text{profile}}/c^2.$$

No equality among these objects is part of their definitions.

Proposition 2.7 (Expectation/Casimir separation). *Membership in a fixed positive-mass irreducible representation and the condition $p_\omega^i = 0$ do not imply*

$$p_\omega^0 = m_{\text{Cas}} c.$$

Indeed a nontrivial symmetric momentum wave packet can satisfy $\langle \mathbf{P} \rangle = 0$ while $\langle H \rangle > m_{\text{Cas}} c^2$.

Proof. On the fixed-mass representation,

$$H = \sqrt{m_{\text{Cas}}^2 c^4 + c^2 \mathbf{P}^2}.$$

For a normalized state with a symmetric nonzero momentum distribution, $\langle \mathbf{P} \rangle = 0$, while strict convexity of the positive-energy dispersion gives

$$\langle H \rangle = \int \sqrt{m_{\text{Cas}}^2 c^4 + c^2 |\mathbf{p}|^2} |\psi(\mathbf{p})|^2 d\mu > m_{\text{Cas}} c^2.$$

The operator identity for $-\hat{P}^2$ therefore does not become an identity for $-\langle \hat{P} \rangle^2$. □

Definition 2.8 (Structured mass-bearing presentation). A structured Module B presentation is

$$\hat{x} = (x, \mathfrak{a}_x, \mathfrak{w}_x, \mathfrak{o}_x, \hat{\mathbf{T}}_x, \mathfrak{p}_x, \mathfrak{n}_x, \mathfrak{f}_x, \mathfrak{s}_x, \mathfrak{c}_x, \mathfrak{t}_x),$$

where:

- $(x, \mathfrak{a}_x, \mathfrak{w}_x) \in \hat{Q}_{\text{mass}}$ carries the accepted Module A source attachment;
- $\mathfrak{o}_x$ is a complete normalized-ray operator/source attachment certificate: it contains one prequotient family of accepted Module A source presentations on $\mathbb{P}_1(\mathfrak{D}_x)$, the associated concrete source distributions on the fixed representative envelope, their norm-squared homogeneous vector extension, and the quadratic polarization certificate; $\hat{\mathbf{T}}_x$ is the unique operator-valued distribution reconstructed from that homogeneous extension;

- $(x, \mathfrak{a}_x, \mathfrak{w}_x, \mathfrak{o}_x, \hat{\mathbf{T}}_x) \in \hat{Q}_{\text{mass}}^{\text{op}}$;
- $\mathfrak{p}_x$ is the full Poincaré representation and stable-sector certificate on the same Hilbert carrier and common domain used by $\mathfrak{o}_x$;
- $\mathfrak{n}_x$ is the current/charge/generator matching certificate for the charge obtained from the attached operator $\hat{\mathbf{T}}_x$;
- $\mathfrak{f}_x$ is the elastic total-stress form-factor certificate for the same attached operator $\hat{\mathbf{T}}_x$;
- $\mathfrak{s}_x$ is the stationary localized profile certificate when the Laue/far-field subclass is invoked;
- $\mathfrak{c}_x$ is the controlled Wigner contraction certificate;
- $\mathfrak{k}_x$ is the calibration and lawful-transport certificate.

Definition 2.9 (Matching equivalence). Two structured presentations are matching-equivalent, written $\hat{x} \sim_{\text{match}} \hat{y}$, when lawful upstream redescription transports every listed certificate, the normalized-ray Module A source-presentation family, its coherent prequotient representative family and homogeneous extension, attached operator tensor, full translation representation, operator/source improvement class, stable mass sector, zero-transfer form factor, contraction limit, and calibration line to the corresponding data of the other presentation.

Remark 2.10. Matching equivalence is intentionally finer than equality in Q_{mass} . A base response class may admit distinct boundary, sector, form-factor, or contraction data. Fibre uniqueness is not assumed here; the module proves that every certified matching presentation produces the same intrinsic proportionality theorem.

3 Imported theorem-use packages and source locks

This module imports only the exact theorem roles listed below. The local project contains hash-locked copies of the principal AASC artifacts. Standard representation and form-factor results are used only after the upstream roles and accepted source object are fixed.

3.1 Mass and inertia role

Mass, Inertia, and Persistence Compression supplies

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D, \quad \iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{\text{tr}}}.$$

Module B uses only the fact that inertial standing is a transport-facing restriction of an already admitted deformation-response role [1]. No equality with a metric source coordinate is imported.

3.2 Schrödinger inertial representation

The Schrödinger normal-form theorem supplies, after an inertial/action scale has standing,

$$H = -\frac{\hbar^2}{2m}\Delta + M_V$$

in the closed scalar nonrelativistic envelope [2]. Module B must construct which stable-sector charge occupies the denominator.

3.3 Accepted Module A

Accepted Module A supplies [3]:

1. the genuine source-bearing subtype $Q_{\text{resp}}^{\text{src}} \subseteq Q_{\text{resp}}$;
2. the structured source quotient $\widehat{Q}_{\text{src}}$ and forgetful map F ;
3. the quotient-faithful distributional source map q_g ;
4. the exact Ward/divergence correspondence;
5. finite completed-composite closure with exact term coverage and internal-exchange cancellation;
6. compatibility with a Hilbert tensor on the independently action-presentable subclass.

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3.4 Poincaré, stress-current, and stable-sector package

Spacetime Symmetry as Domain Anchor supplies the fixed Minkowski/Wightman target with a strongly continuous Poincaré action, self-adjoint generators on a common invariant domain, stable massive sectors, and an explicitly admitted local stress-current/charge package [4]. Module B does not infer a current merely from covariance. The corrected charge comparison is made on the full irreducible vacuum/reference representation and only then restricted to the stable spectral sector.

3.5 Total energy–momentum form factors

For a stable spin-zero particle the matrix elements of the conserved total energy–momentum tensor have two form factors [5, 6, 7]. Translation charge normalization forces $A(0) = 1$; the independent D -term records internal stress but does not alter the zero-transfer momentum charge. Module B re-proves the normalization from the global charge theorem in its own conventions. This form-factor theorem is the corrected bridge between the stable-sector Casimir and the active soft-gravitational source. It replaces the rejected identification of the Casimir with an arbitrary state’s expectation four-momentum.

3.6 Einstein first response

The Einstein projection theorem supplies

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

and the metric-facing source role [8]. Module B uses only its linearized stationary first-response consequence for the accepted total source.

3.7 Wigner, Bargmann, and contraction results

Wigner’s classification fixes the standard positive-mass spin-zero one-particle representation; Bargmann’s theorem identifies nonrelativistic mass as the central charge; and Inönü–Wigner contraction supplies the group-theoretic limit [9, 10, 11]. The present manuscript does not cite these facts as an existence substitute: it writes the scalar momentum-space generators, constructs the unitaries, proves common-core convergence, and exhibits the free Klein–Gordon sector as a full inhabitant. Essential self-adjointness is controlled by the Stone generators and the standard complete-flow criterion for first-order generators [12].

Proposition 3.1 (Import noncollapse). *The upstream response class, normalized-ray Module A source-presentation family, coherent concrete representative family, homogeneous vector extension, reconstructed operator total EMT, global translation charge, stable-sector Casimir, zero-transfer form factor, Breit source profile, active metric monopole, Bargmann central charge, and Schrödinger kinetic coefficient are distinct theorem objects. The operator/source attachment theorem of section 4 is the first required bridge; the later charge, form-factor, and contraction bridges follow only after it.*

4 Operator-presentable Module A sources from normalized rays

Accepted Module A constructs a concrete metric-source distribution from an attached operational response presentation and only afterward passes to lawful source equivalence. A Poincaré form-factor theorem, by contrast, requires one operator-valued distribution on a common quantum domain. These are not automatically the same object. The present section constructs the exact bridge without conflating a physical projective state with a non-normalized vector argument of a quadratic form.

The correction has three stages:

$$\begin{aligned} \text{normalized ray source presentations} &\implies \text{concrete source representatives} \\ &\implies \text{homogeneous vector-weight extension} \\ &\implies \text{operator by complex polarization.} \end{aligned}$$

The first arrow is physical-state evaluation. The second is the accepted Module A operational-to-distributional theorem on the prequotient source presentation. The third is an algebraic extension used only to reconstruct matrix elements. It is not itself a map from physical ray states.

4.1 Prequotient Module A source presentations

Fix a full reference Hilbert space $\mathcal{H}_{\text{ref}}$ and a dense complex locally convex domain

$$\mathfrak{D}_x \subset \mathcal{H}_{\text{ref}}$$

that is invariant under the lawful Poincaré and field operations used below. Its continuous anti-dual is $\mathfrak{D}'_x$. The graph topology on $\mathfrak{D}_x$ and the strong anti-dual topology on $\mathfrak{D}'_x$ are part of the certificate.

Let

$$\text{SrcPres}_g^A := \{(p, \mathfrak{a}_p, \mathfrak{w}_p) : \text{SrcBear}_g(p, \mathfrak{a}_p, \mathfrak{w}_p)\}$$

denote the accepted Module A source-bearing presentations *before* lawful source quotienting. Accepted

Module A supplies two maps on this prequotient class:

$$\pi_A : \text{SrcPres}_g^A \longrightarrow \hat{Q}_{\text{src}}, \quad \mathbb{T}_A : \text{SrcPres}_g^A \longrightarrow \text{Src}_g^{\text{dist}},$$

where π_A is the lawful presentation quotient and $\mathbb{T}_A(p, \mathfrak{a}_p, \mathfrak{w}_p)$ is the unique concrete symmetric tensor-density distribution representing the attached metric-response witness. By construction,

$$[\mathbb{T}_A(s)]_{\simeq_{\text{law}}} = q_g(\pi_A(s)) \quad (s \in \text{SrcPres}_g^A). \quad (1)$$

Thus a concrete source representative is obtained from the operational source presentation itself; it is not selected afterward from the quotient class.

4.2 Normalized ray-state source presentations

Define the normalized projective state space on the common core by

$$\mathbb{P}_1(\mathfrak{D}_x) := \{\psi \in \mathfrak{D}_x : \|\psi\| = 1\} / U(1).$$

The zero vector is not an element of $\mathbb{P}_1(\mathfrak{D}_x)$. It enters only later as the zero of the homogeneous algebraic extension.

Definition 4.1 (Normalized ray-state Module A source-presentation family). Let $\hat{x} \in \hat{Q}_{\text{mass}}$. A normalized ray-state source-presentation family over $\hat{x}$ is a map

$$\tilde{\mathfrak{s}}_x : \mathbb{P}_1(\mathfrak{D}_x) \longrightarrow \text{SrcPres}_g^A, \quad [\psi] \longmapsto \tilde{s}_{x,[\psi]},$$

with the following properties.

1. **Common upstream role.** Every $F(\pi_A(\tilde{s}_{x,[\psi]}))$ is the normalized-state evaluation of the same mass-bearing response role $F(\hat{x})$. Changing the ray does not change the upstream role, projection anchors, or source-attachment type.
2. **Operational formation before quotient.** Each $\tilde{s}_{x,[\psi]}$ is the output of the one fixed state-evaluation protocol in the accepted Module A source presentation. The family is not a section chosen after applying q_g .
3. **Fixed representative envelope.** All values use the same spacetime, metric anchor, boundary class, regularity class, normalization convention, and improvement representative fixed by the matching certificate. State-dependent changes of source gauge, improvement, or boundary representative are forbidden.
4. **Ray totality and projective fidelity.** The family is defined for every normalized ray. Its domain is already quotiented by phase, so no separate scalar-invariance clause is needed.
5. **Lawful covariance.** Whenever a lawful unitary symmetry $U(g)$ preserves the common core,

$$\tilde{\mathfrak{s}}_x([U(g)\psi]) = g_*\tilde{\mathfrak{s}}_x([\psi])$$

in the fixed source-presentation envelope.

6. **Ray-source completeness.** Every normalized physical ray state admitted on the common domain has exactly the source presentation produced by this protocol family; no independently

standing normalized-ray source is omitted.

The associated quotient-valued ray family is derived, not separately selected:

$$\mathfrak{s}_x^{\text{ray}} := \pi_{\mathbf{A}} \circ \tilde{\mathfrak{s}}_x : \mathbb{P}_1(\mathfrak{D}_x) \longrightarrow \widehat{Q}_{\text{src}}.$$

The concrete normalized ray-state source distribution is

$$\mathbf{T}_{x,[\psi]}^{\text{ray}} := \mathbb{T}_{\mathbf{A}}(\tilde{\mathfrak{s}}_{x,[\psi]}).$$

By (1),

$$[\mathbf{T}_{x,[\psi]}^{\text{ray}}]_{\simeq_{\text{law}}} = q_g(\mathfrak{s}_x^{\text{ray}}([\psi])). \quad (2)$$

Equation (2) is the coherent representative-lift theorem needed for polarization: all representatives are formed on one fixed carrier by one prequotient operational protocol before lawful quotienting.

4.3 Homogeneous vector-weight extension

Definition 4.2 (Homogeneous extension of the normalized ray source). For $\psi \in \mathfrak{D}_x$, define

$$\mathbf{T}_{x,\psi}^{\sharp} := \begin{cases} \|\psi\|^2 \mathbf{T}_{x,[\psi]}^{\text{ray}}, & \psi \neq 0, \\ 0, & \psi = 0. \end{cases}$$

For $h \in \Gamma_c(S^2 T^* M)$, set

$$q_{x,h}^{\sharp}(\psi) := \langle \mathbf{T}_{x,\psi}^{\sharp}, h \rangle.$$

The map $\psi \mapsto \mathbf{T}_{x,\psi}^{\sharp}$ is an algebraic vector-weight extension of the physical ray-state source family. It is not itself a physical-state assignment on projective Hilbert space.

Proposition 4.3 (Well-defined projective-to-homogeneous passage). *The extension of definition 4.2 is independent of the unit representative of the normalized ray and satisfies*

$$\mathbf{T}_{x,\lambda\psi}^{\sharp} = |\lambda|^2 \mathbf{T}_{x,\psi}^{\sharp}, \quad q_{x,h}^{\sharp}(\lambda\psi) = |\lambda|^2 q_{x,h}^{\sharp}(\psi)$$

for every $\lambda \in \mathbb{C}$. For $\|\psi\| = 1$,

$$\mathbf{T}_{x,\psi}^{\sharp} = \mathbf{T}_{x,[\psi]}^{\text{ray}}.$$

Proof. If ψ and $e^{i\theta}\psi$ are unit representatives of the same ray, the value of $\mathbf{T}_{x,[\psi]}^{\text{ray}}$ is unchanged because the family is defined on $\mathbb{P}_1(\mathfrak{D}_x)$. If $\lambda \neq 0$, then $[\lambda\psi/\|\lambda\psi\|] = [\psi/\|\psi\|]$ and $\|\lambda\psi\|^2 = |\lambda|^2 \|\psi\|^2$. The zero case is immediate. $\square$

Remark 4.4 (Why projective standing and quadraticity no longer conflict). The physical source is $\mathbf{T}_{x,[\psi]}^{\text{ray}}$, indexed by normalized rays and therefore invariant under all nonzero scalar changes of representative. The degree-two law belongs only to $\mathbf{T}_{x,\psi}^{\sharp}$, an auxiliary extension on vector weights used to reconstruct an operator quadratic form. The two objects have different domains and are not identified.

Definition 4.5 (Quadratic ray-source certificate). A normalized ray-state source-presentation family carries a quadratic ray-source certificate when, for every real compact metric test tensor h , the homogeneous functional $q_{x,h}^{\sharp} : \mathfrak{D}_x \rightarrow \mathbb{R}$ satisfies:

1. **parallelogram identity:**

$$q_{x,h}^\sharp(\phi + \psi) + q_{x,h}^\sharp(\phi - \psi) = 2q_{x,h}^\sharp(\phi) + 2q_{x,h}^\sharp(\psi);$$

2. **joint continuity:** the polarized form is jointly continuous in (ϕ, ψ, h) for the graph topology on $\mathfrak{D}_x$ and the LF topology on the metric test space;
3. **homogeneous-source completeness:** every diagonal vector-weight source evaluation admitted on the declared common domain is the extension of the normalized ray-state family above; no independent quadratic source family is appended.

Degree-two homogeneity and phase behavior are conclusions of proposition 4.3, not additional state-equivalence assumptions. The certificate contains no operator energy–momentum tensor as input.

4.4 Polarization reconstruction

We use the convention that the reconstructed form is conjugate-linear in its first argument and linear in its second. For fixed h , define

$$b_{x,h}(\phi, \psi) := \frac{1}{4} \left[q_{x,h}^\sharp(\phi + \psi) - q_{x,h}^\sharp(\phi - \psi) - iq_{x,h}^\sharp(\phi + i\psi) + iq_{x,h}^\sharp(\phi - i\psi) \right]. \quad (3)$$

Theorem 4.6 (Operator-source polarization theorem). *Let $\hat{x} \in \hat{Q}_{\text{mass}}$ carry a normalized ray-state Module A source-presentation family, its concrete prequotient representative family, the homogeneous extension of definition 4.2, and a quadratic ray-source certificate. Then there exists a unique continuous linear map*

$$\hat{\mathbf{T}}_x : \Gamma_c(S^2T^*M) \longrightarrow \mathcal{L}(\mathfrak{D}_x, \mathfrak{D}'_x)$$

such that

$$\langle \phi, \hat{\mathbf{T}}_x(h)\psi \rangle = b_{x,h}(\phi, \psi)$$

for all $\phi, \psi \in \mathfrak{D}_x$ and all compact metric variations h . It is Hermitian as a quadratic-form-valued distribution, and

$$\langle \psi, \hat{\mathbf{T}}_x(h)\psi \rangle = q_{x,h}^\sharp(\psi) = \langle \mathbf{T}_{x,\psi}^\sharp, h \rangle.$$

For normalized ψ , this is exactly the concrete accepted Module A ray-state source:

$$\langle \psi, \hat{\mathbf{T}}_x(h)\psi \rangle = \langle \mathbf{T}_{x,[\psi]}^{\text{ray}}, h \rangle.$$

No different operator-valued distribution has the same complete homogeneous extension of the normalized ray source family.

Proof. By proposition 4.3, $q_{x,h}^\sharp$ is homogeneous of degree two. Together with the parallelogram identity, the complex polarization identity (3) yields a Hermitian sesquilinear form with diagonal $q_{x,h}^\sharp$. Joint continuity makes $\phi \mapsto b_{x,h}(\phi, \psi)$ a continuous functional on $\mathfrak{D}_x$ for every ψ , hence an element of $\mathfrak{D}'_x$. Define $\hat{\mathbf{T}}_x(h)\psi$ by this functional. Linearity in h follows from distributional linearity of every concrete Module A source representative and the norm-squared extension. Joint continuity gives continuity of $h \mapsto \hat{\mathbf{T}}_x(h)$ into $\mathcal{L}(\mathfrak{D}_x, \mathfrak{D}'_x)$. If $\hat{\mathbf{T}}_x$ has the same diagonal forms, polarization forces every off-diagonal form to agree. $\square$

Definition 4.7 (Operator/source attachment). Write

$$\text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x)$$

when $\mathfrak{o}_x$ contains the common Hilbert/domain data, the normalized ray-state accepted Module A source-presentation family, the concrete prequotient representative family, the homogeneous vector-weight extension, and the quadratic ray-source certificate, and when $\hat{\mathbf{T}}_x$ is the unique polarization reconstruction of theorem 4.6. The relation is an incidence theorem between accepted Module A ray-state sources and an operator-valued distribution. It is not satisfied by adjoining an unrelated operator current or by choosing independent representatives of source quotient classes state by state.

Corollary 4.8 (No independent operator-current insertion). *If an operator-valued distribution $\hat{S}$ differs from $\hat{\mathbf{T}}_x$ on any matrix element over $\mathfrak{D}_x$, then $\hat{S}$ is not operator-attached to $\hat{x}$. It may describe another source theory, an added tensor load, or a richer target, but it cannot supply the Module B active branch of $\hat{x}$.*

4.5 Operator-presentable mass-bearing subtype

Definition 4.9 (Operator-presentable mass-bearing subtype). The structured operator-presentable subtype is

$$\hat{Q}_{\text{mass}}^{\text{op}} := \left\{ (\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x) : \hat{x} \in \hat{Q}_{\text{mass}}, \text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x) \right\} / \sim_{\text{op}},$$

where $\sim_{\text{op}}$ transports the upstream response, normalized ray-state source-presentation family, its concrete prequotient representatives, the homogeneous extension, the common core, and the reconstructed operator distribution by lawful unitary/source equivalence. The forgetful map

$$F_{\text{op}} : \hat{Q}_{\text{mass}}^{\text{op}} \longrightarrow \hat{Q}_{\text{mass}}$$

is well defined. No claim is made that every source-attached mass-bearing object is operator presentable.

Proposition 4.10 (The operator subtype is non-tautological). *An accepted Module A source attachment with no common quantum domain, no total normalized ray-state source-presentation family, no coherent prequotient representative family, or a homogeneous extension that fails the parallelogram or continuity conditions belongs to $\hat{Q}_{\text{mass}}$ but not to $\hat{Q}_{\text{mass}}^{\text{op}}$.*

Proof. Accepted Module A requires an operational metric-response functional and its distributional source; it does not require a projective quantum state family, a coherent prequotient source realization on one carrier, or diagonal matrix elements of one operator-valued distribution. Failure of any listed condition blocks definition 4.7. $\square$

4.6 Transport of the accepted Module A ledgers

Theorem 4.11 (Support, covariance, Ward, and boundary inheritance). *Suppose $\text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x)$. Then the following normalized ray-state accepted Module A properties lift uniquely to the operator form domain.*

1. *If every concrete normalized ray source has support in the same declared closed envelope K_x , then $\hat{\mathbf{T}}_x(h) = 0$ whenever $\text{supp } h \cap K_x = \emptyset$.*

2. If lawful redescription χ is implemented by $U_x(\chi)$ and the prequotient ray source family satisfies Module A naturality, then

$$U_x(\chi) \hat{\mathbf{T}}_x(h) U_x(\chi)^{-1} = \hat{\mathbf{T}}_x(\chi_* h)$$

as forms on the transported common domain.

3. If the normalized ray-state Module A exchange currents possess the same coherent prequotient lift and their homogeneous extensions form a jointly continuous quadratic family, they reconstruct a unique operator-valued current $\hat{\mathbf{J}}_x$, and

$$\hat{\mathbf{T}}_x(\mathcal{L}_X g) = -2\hat{\mathbf{J}}_x(X)$$

as a quadratic-form identity.

4. Ray-state regularity, boundary, and lawful-improvement certificates transport to the reconstructed form distribution; an improvement invisible in every concrete normalized ray source is invisible in every matrix element.

Proof. First extend each normalized ray-state equality by the norm-squared construction of definition 4.2; the resulting identities hold on every diagonal vector. Apply the complex polarization identity to those homogeneous diagonal equalities. A Hermitian sesquilinear form that vanishes, covaries, satisfies a Ward identity, or differs by an improvement on every diagonal vector has the same property on every off-diagonal pair. Uniqueness follows from theorem 4.6. $\square$

Corollary 4.12 (Integrated charge-form compatibility). *Whenever the hypersurface charge of the concrete normalized ray-state Module A source family is defined on the common matching domain, write*

$$P_{\text{src},x,[\psi]}^\mu$$

for the charge of $\mathbf{T}_{x,[\psi]}^{\text{ray}}$. Then the attached operator charge $\hat{Q}_x^\mu$ satisfies

$$\langle \psi, \hat{Q}_x^\mu \psi \rangle = \|\psi\|^2 P_{\text{src},x,[\psi/\|\psi\|]}^\mu \quad (\psi \neq 0),$$

and the diagonal is zero at $\psi = 0$. In particular, for normalized ψ ,

$$\langle \psi, \hat{Q}_x^\mu \psi \rangle = P_{\text{src},x,[\psi]}^\mu.$$

All off-diagonal charge forms are fixed by polarization. Thus the charge-generation theorem of Part II may use only the charge obtained from $\hat{\mathbf{T}}_x$.

Scope of the bridge. The theorem constructs an operator realization only for $\hat{Q}_{\text{mass}}^{\text{op}}$. It does not promote all Module A sources to operator currents. It does not treat a non-normalized vector as a new physical quantum state. It does not choose source representatives state by state after quotienting. The physical input is a normalized ray-state source presentation; the non-normalized vector dependence is only the homogeneous extension required for polarization.

5 The corrected Poincaré–Noether/form-factor/Breit/Bargmann envelope

Definition 5.1 (Corrected matching object). An object of $\mathbf{Match}_D^{\text{mass}}$ is a structured operator-presentable source-attached mass-bearing presentation whose first five coordinates define an element of $\hat{Q}_{\text{mass}}^{\text{op}}$, carrying the following exact clauses.

- M1. Upstream type.** $(x, \mathfrak{a}_x, \mathfrak{w}_x) \in \hat{Q}_{\text{mass}}$, with the accepted Module A source attachment.
- M2. Operator/source attachment.** A common dense invariant domain $\mathfrak{D}_x$, a total normalized ray-state family of accepted Module A source presentations formed before quotienting, one coherent concrete representative family on the fixed source envelope, its norm-squared homogeneous vector extension, and the quadratic/polarization certificate of section 4 reconstruct one unique operator-valued distribution $\hat{\mathbf{T}}_x$. Thus

$$(x, \mathfrak{a}_x, \mathfrak{w}_x, \mathfrak{o}_x, \hat{\mathbf{T}}_x) \in \hat{Q}_{\text{mass}}^{\text{op}}.$$
- M3. Full reference representation and source regularity.** A strongly continuous positive-energy Poincaré representation $(\mathcal{H}_{\text{ref}}, \pi, \Omega, U)$ is fixed on the same Hilbert carrier and preserves $\mathfrak{D}_x$; $\pi(\mathcal{A})''$ acts irreducibly and Ω is the unique invariant reference vector used for the zero baseline. The attached operator source and its transition family are tempered and admit the hypersurface, Fourier, and first-response pairings used here.
- M4. Global attached energy–momentum charge package.** The operator $\hat{T}^{\mu\nu}$ used in every later clause is exactly the reconstructed $\hat{\mathbf{T}}_x^{\mu\nu}$. Its improvement-controlled whole-space charges have unitary groups implementing the same translations as U on the full reference representation.
- M5. Stable scalar sector.** A nonzero spectral subspace $\mathcal{H}_{m_{\text{Cas}}}$ carries a stable positive-mass spin-zero Wigner representation with one Casimir value $m_{\text{Cas}} > 0$.
- M6. Attached total-EMT form-factor certificate.** Covariantly normalized generalized states $|p\rangle$ exist on the mass shell; the matrix element of the same attached operator $\hat{\mathbf{T}}_x^{\mu\nu}$ is a tempered distribution, has the spin-zero two-form-factor decomposition, and is continuous at zero momentum transfer. No value of $A(0)$ is included in this clause.
- M7. Breit source certificate.** An even Schwartz regulator with value one at the origin produces a stationary conserved Breit source profile with the moments and boundary behavior required by Laue closure and the linearized far-field solution. The profile is a transition-source distribution, not the expectation value of a normal localized sharp-momentum state.
- M8. Einstein first response.** The metric image admits the stationary weak-field harmonic- gauge expansion about Minkowski space with fixed coupling κ .
- M9. Scalar Wigner contraction.** The stable sector is unitarily equivalent to the standard scalar Wigner representation. The explicit common-core contraction of Part IV is part of the certificate; no bare Boolean substitutes for its analytic construction.
- M10. Calibration and transport.** The active and inertial scalar lines have one fixed source calibration, one fixed inertial calibration, and flat lawful relative calibration transport.

Definition 5.2 (Lawful matching morphism). A morphism in $\mathbf{Match}_D^{\text{mass}}$ preserves the upstream response class, the normalized-ray Module A source-presentation family, its coherent prequotient representatives and homogeneous extension, operator/source attachment, common core, full reference representation up to unitary equivalence, attached total-EMT improvement class, stable scalar spectral sector, form-factor family, Breit regulator class, explicit contraction, and calibrations. It may change coordinates, basis, gauge representative, form-factor notation, or improvement representative only through a certified witness-neutral equivalence.

Definition 5.3 (Calibration-connected envelope). The wider category $\mathbf{Env}_D^{\text{mass}}$ contains mass-bearing objects connected to $\mathbf{Match}_D^{\text{mass}}$ by lawful response, source, inertial, and calibration transports. Every connected component intersects $\mathbf{Match}_D^{\text{mass}}$, and relative calibration transport has trivial holonomy. Nontrivial relative scaling is new calibration tensor content and therefore a different target.

Proposition 5.4 (The matching clauses do not contain the conclusion). *The certificate fixes where all constructions are well typed. It does not assert: equality of the global total-EMT charge with the Stone generator; $A(0) = 1$; equality of the active monopole with the Casimir; existence of the Bargmann central term after contraction; or a universal factor. Those are proved in Parts II–V.*

The theorem is an exact overlap result for stable spin-zero sectors. It does not assign a global mass to generic curved spacetime, unstable resonances, massless sectors, radiating systems, or an arbitrary normal wave packet. In particular, the invariant norm of a state's mean four-momentum is not used as the stable-sector mass.

Part II

Common Translation Charge

6 The accepted total source charge

Fix $\hat{x} \in \mathbf{Match}_D^{\text{mass}}$. Accepted Module A supplies, through M2, one concrete normalized ray-state family of completed source distributions on the fixed matching carrier. Every member contains each certified constituent, binding, mediator, interaction, internal-stress, restoration, transport, baseline, and boundary contribution exactly once. The norm-squared homogeneous extension of that family reconstructs the unique attached operator-valued distribution $\hat{\mathbf{T}}_x^{\mu\nu}$ by theorem 4.6.

Definition 6.1 (Normalized ray-state translation charge). On a Minkowski matching region with signature $(-+++)$, let $\mathbf{T}_{x,[\psi]}^{\text{ray}}$ be the concrete accepted Module A source distribution of a normalized ray $[\psi] \in \mathbb{P}_1(\mathfrak{D}_x)$. Define

$$P_{\text{src},x,[\psi]}^{\mu}(\Sigma) := \frac{1}{c} \int_{\Sigma} (\mathbf{T}_{x,[\psi]}^{\text{ray}})^{\mu\nu} d\Sigma_{\nu}^{(+)}$$

whenever the accepted distributional pairing and whole-space limit exist. For an arbitrary vector

$\psi \in \mathfrak{D}_x$, define the homogeneous charge form

$$P_{\text{src},x,\psi}^{\sharp\mu}(\Sigma) := \begin{cases} \|\psi\|^2 P_{\text{src},x,[\psi/\|\psi\|]}^{\mu}(\Sigma), & \psi \neq 0, \\ 0, & \psi = 0. \end{cases}$$

The first object is a physical normalized-state source charge. The second is its vector-weight extension and is used only as the diagonal form of an operator charge.

Lemma 6.2 (Hypersurface independence). *If the completed normalized ray-state tensor is closed, $\partial_\nu(\mathbf{T}_{x,[\psi]}^{\text{ray}})^{\mu\nu} = 0$, and the certified spatial flux vanishes, then $P_{\text{src},x,[\psi]}^{\mu}(\Sigma)$ is independent of the Cauchy hyperplane. The same is therefore true of its homogeneous extension.*

Proof. Pair distributional conservation with a cutoff translation field on the slab between two hyperplanes. Stokes' theorem gives the difference of the two charges plus the timelike boundary flux. The zero-flux certificate removes the latter in the whole-space limit. Multiplication by the constant $\|\psi\|^2$ preserves the equality for the homogeneous extension. $\square$

Proposition 6.3 (Complete-composite charge). *Every normalized ray-state charge is formed from an accepted Module A total tensor and therefore includes the full exact source registry. It is not a sum of bare constituent mass labels. The same statement holds for the homogeneous charge form.*

Proof. Module A proves exact term coverage, nonduplication, and cancellation of every balanced internal exchange column. Linearity of the hypersurface pairing then gives the total charge. Missing or duplicated terms violate the accepted certificate before Module B begins. $\square$

Proposition 6.4 (State/operator charge incidence). *Let the hypersurface integrals be defined on the common matching domain. The operator charge forms reconstructed from $\hat{\mathbf{T}}_x$ satisfy*

$$\langle \psi, \hat{Q}_x^{\mu}(\Sigma) \psi \rangle = P_{\text{src},x,\psi}^{\sharp\mu}(\Sigma)$$

for every $\psi \in \mathfrak{D}_x$. In particular, for normalized ψ ,

$$\langle \psi, \hat{Q}_x^{\mu}(\Sigma) \psi \rangle = P_{\text{src},x,[\psi]}^{\mu}(\Sigma).$$

All off-diagonal matrix elements are uniquely fixed by complex polarization. Hence the global charge package of M_4 is the operator realization of the accepted Module A normalized ray-state source charges of M_1 – M_2 .

Proof. The diagonal equality is corollary 4.12. The hypersurface integral is linear on the admitted distributional class, so integration commutes with the norm-squared homogeneous extension and with polarization. The complex polarization identity then fixes every off-diagonal charge form. No independent charge operator can share all normalized ray-state accepted source charges and their homogeneous extension while differing on the common core. $\square$

Definition 6.5 (Attached improvement equivalence). Two attached operator total tensors are charge-equivalent when

$$\hat{\mathbf{T}}_x'^{\mu\nu} = \hat{\mathbf{T}}_x^{\mu\nu} + \partial_\lambda \hat{B}_x^{\lambda\mu\nu}, \quad \hat{B}_x^{\lambda\mu\nu} = -\hat{B}_x^{\mu\lambda\nu},$$

and the fixed boundary certificate kills the resulting surface charge.

Lemma 6.6 (Improvement invariance). *Charge-equivalent attached operator tensors have the same global translation charge and the same zero-transfer form-factor normalization.*

Proof. The charge difference is a boundary integral. In momentum space the improvement contribution is proportional to the momentum transfer, so it vanishes at $\Delta = 0$. Both conclusions use the same fixed boundary/improvement certificate. $\square$

Remark 6.7. This section constructs the global charge only when the no-flux and integrability certificates hold. No generic curved-spacetime energy is inferred. A non-normalized vector is never treated as a new physical source state; it appears only as the argument of the homogeneous operator quadratic form.

7 Global charge-generator equality before spectral restriction

Let $(\mathcal{H}_{\text{ref}}, \pi, \Omega, U)$ be the full reference representation from M3. Write

$$U(a) = \exp\left(-\frac{i}{\hbar} a_\mu \hat{P}^\mu\right).$$

The charges obtained by integrating the attached operator tensor $\hat{\mathbf{T}}_x^{\mu\nu}$ have self-adjoint closures $\hat{Q}_x^\mu$ and unitary group

$$V(a) = \exp\left(-\frac{i}{\hbar} a_\mu \hat{Q}_x^\mu\right).$$

Assumption 7.1 (Exact global charge-generation certificate). On the full reference representation, the group $V(a)$ generated by the attached source charge and $U(a)$ implement the same translation automorphisms of $\pi(\mathcal{A})''$, are strongly continuous, preserve the common Gårding domain, and leave the invariant reference vector fixed. The improvement representative is fixed only up to lemma 6.6.

Lemma 7.2 (Commutant phase). *For every translation a , the unitary $W(a) := V(a)^*U(a)$ belongs to the commutant $\pi(\mathcal{A})'$ and is therefore scalar.*

Proof. Because the two groups implement the same automorphism, $W(a)\pi(A)W(a)^* = \pi(A)$ for every observable A . Thus $W(a)$ lies in the commutant. M3 fixes an irreducible reference representation, so Schur's lemma gives $W(a) = e^{i\chi(a)}I$. $\square$

Theorem 7.3 (Global source-charge/generator matching). *The invariant-reference baseline removes the commutant phase:*

$$V(a) = U(a) \quad \text{for all } a, \quad \hat{Q}_x^\mu = \hat{P}^\mu.$$

Proof. Both groups leave Ω fixed. Hence $e^{i\chi(a)}\Omega = W(a)\Omega = \Omega$, so the continuous character is identically one. Equality of strongly continuous unitary groups gives equality of their self-adjoint generators by Stone's theorem. By proposition 6.4, the generators $\hat{Q}_x^\mu$ are the operator realization of the accepted Module A state-source charges of the same upstream object. The proof is global; it does not place the vacuum or the local observable algebra inside a one-particle subspace. $\square$

Corollary 7.4 (Restriction to the stable sector). *Let $E_{m_{\text{Cas}}}$ be the joint spectral projection onto the stable positive-mass spin-zero sector. Then*

$$E_{m_{\text{Cas}}} \hat{Q}_x^\mu E_{m_{\text{Cas}}} = E_{m_{\text{Cas}}} \hat{P}^\mu E_{m_{\text{Cas}}}.$$

Proof. The equality holds on the full representation. The spectral projection is a Borel function of the commuting translation generators, so restriction preserves it. $\square$

For a normalized wave packet, the expectation identity $\omega(\hat{Q}_x^\mu) = \omega(\hat{P}^\mu)$ is valid, but it does *not* imply $-\omega(P)^2 = \omega(-P^2)$. No later theorem uses that invalid inference.

8 Stable-sector Poincaré Casimir

Let $E_{m_{\text{Cas}}}$ be the joint spectral projection onto the stable positive-mass spin-zero sector fixed by M5. On that sector,

$$-\eta_{\mu\nu} \hat{P}^\mu \hat{P}^\nu = m_{\text{Cas}}^2 c^2 I, \quad m_{\text{Cas}} > 0.$$

This is an operator and spectral statement. It is not the invariant norm of the expectation four-momentum of an arbitrary state.

Definition 8.1 (Casimir mass line). For a matching object x , define the relativistic charge line

$$\mathcal{L}_{D,x}^P := \mathbb{R}_{>0} m_{\text{Cas}}(x)$$

with the orientation fixed by positive energy. Generalized momentum states in the stable sector satisfy

$$-\eta_{\mu\nu} p^\mu p^\nu = m_{\text{Cas}}^2 c^2, \quad p^0 > 0.$$

Proposition 8.2 (Casimir/expectation separation). *For a normalized nontrivial wave packet ψ in the same sector,*

$$-\eta_{\mu\nu} \langle \hat{P}^\mu \rangle_\psi \langle \hat{P}^\nu \rangle_\psi$$

need not equal $m_{\text{Cas}}^2 c^2$. In particular, a symmetric mean-rest packet can have $\langle \mathbf{P} \rangle_\psi = 0$ and $\langle H \rangle_\psi > m_{\text{Cas}} c^2$.

Proof. On the sector,

$$H = \sqrt{m_{\text{Cas}}^2 c^4 + c^2 \mathbf{P}^2}.$$

For a normalized state with nonzero symmetric momentum distribution, the mean spatial momentum vanishes, whereas the positive-energy dispersion is strictly larger than $m_{\text{Cas}} c^2$ away from $\mathbf{p} = 0$. Integration therefore gives the strict energy inequality. The operator Casimir identity remains unchanged. $\square$

The intrinsic common charge used below is the stable-sector Casimir together with the exact total energy–momentum Ward normalization. No selected-state expectation energy is substituted for it.

9 Exact zero-transfer total-EMT form-factor normalization

The active branch is not obtained from the norm of a selected state’s expectation four-momentum. It is obtained from the complete total energy–momentum tensor at zero momentum transfer, where the translation Ward identity fixes the intrinsic scalar monopole.

9.1 Generalized stable-sector matrix elements

For the form-factor calculation set $c = \hbar = 1$; physical units are restored in sections 11 and 12. Let $|p\rangle$ denote covariantly normalized generalized momentum states in the stable scalar sector:

$$\langle p'|p\rangle = 2p^0(2\pi)^3\delta^{(3)}(\mathbf{p}' - \mathbf{p}), \quad -p^2 = m_{\text{Cas}}^2, \quad p^0 > 0.$$

These are rigged-Hilbert-space spectral distributions, not normalizable localized states.

Definition 9.1 (Total spin-zero stress-tensor form factors). Set

$$\bar{P} := \frac{p' + p}{2}, \quad \Delta := p' - p, \quad t := \Delta^2.$$

The M6 certificate requires the matrix element of the operator/source-attached total EMT to be a tempered distribution with scalar coefficients regular at $t = 0$, and to admit the decomposition

$$\Theta_x^{\mu\nu}(p', p) := \langle p' | \hat{\mathbf{T}}_x^{\mu\nu}(0) | p \rangle = 2\bar{P}^\mu \bar{P}^\nu A_x(t) + \frac{1}{2}(\Delta^\mu \Delta^\nu - \eta^{\mu\nu} \Delta^2) D_x(t).$$

The value of $A_x(0)$ is not included in the certificate.

Proposition 9.2 (Form-factor/source incidence). *Every generalized matrix element $\Theta_x^{\mu\nu}(p', p)$ used in this section is a matrix element of the unique operator $\hat{\mathbf{T}}_x$ reconstructed from the norm-squared homogeneous extension of the concrete accepted Module A normalized-ray source-presentation family of the same upstream object. For normalized vectors, its diagonal quadratic forms agree exactly with those concrete Module A ray-state sources; for arbitrary vectors, they agree with the homogeneous extension. Its off-diagonal generalized matrix elements are the resulting polarization/rigged-space continuation. No independently supplied total EMT is admitted.*

Proof. The common-domain matrix elements are fixed by theorem 4.6. M3–M6 extend the same attached form distribution to the Poincaré-invariant rigged spectral domain and require tempered continuity. Uniqueness of polarization and distributional continuation on the stated dense core identifies the matrix-element family with the attached operator. An unrelated operator violates M2 and corollary 4.8. $\square$

Proposition 9.3 (Completeness of the scalar decomposition). *For a conserved symmetric parity-even total stress tensor between equal-mass spin-zero states, the two tensors displayed in definition 9.1 exhaust the elastic matrix element.*

Proof. Lorentz covariance allows symmetric tensors made from $\bar{P}$, Δ , and η . Equal on-shell masses give $\bar{P} \cdot \Delta = 0$, while conservation removes longitudinal combinations. Parity and spin zero exclude pseudotensor and spin structures. The two remaining transverse symmetric tensors form the usual spin-zero basis [5, 6, 7]. $\square$

Lemma 9.4 (Translation-charge matrix element). *The full-representation equality $\hat{Q}_x^\nu = \hat{P}^\nu$ from theorem 7.3 implies*

$$\Theta_x^{0\nu}(p, p) = 2p^0 p^\nu.$$

Proof. Translation covariance gives

$$\langle p' | \hat{\mathbf{T}}_x^{0\nu}(x) | p \rangle = e^{i(p' - p) \cdot x} \Theta_x^{0\nu}(p', p).$$

Integration over a constant-time hyperplane produces the spatial delta distribution. The global charge equality then gives

$$\langle p' | \hat{Q}_x^\nu | p \rangle = \langle p' | \hat{P}^\nu | p \rangle = 2p^0 p^\nu (2\pi)^3 \delta^{(3)}(\mathbf{p}' - \mathbf{p}).$$

Equality of the distributional coefficients on the diagonal yields the claim. $\square$

Theorem 9.5 (Zero-transfer translation Ward normalization). *For every object of $\mathbf{Match}_D^{\text{mass}}$,*

$$A_x(0) = 1.$$

Proof. At $p' = p$, $\Delta = 0$ and $\bar{P} = p$. The decomposition becomes

$$\Theta_x^{0\nu}(p, p) = 2p^0 p^\nu A_x(0).$$

Comparison with lemma 9.4 and $p^0 > 0$ fixes the coefficient to one. $\square$

Corollary 9.6 (Intrinsic zero-transfer source mass). *In the rest channel $\bar{p}^\mu = (m_{\text{Cas}}, \mathbf{0})$, define*

$$E_{\text{ff}}(x) := \lim_{\substack{\Delta \rightarrow 0 \\ \Delta^0 = 0}} \frac{\Theta_x^{00}(\bar{p} + \Delta/2, \bar{p} - \Delta/2)}{2\bar{p}^0}, \quad m_{\text{ff}}(x) := E_{\text{ff}}(x)$$

in natural units. Then

$$E_{\text{ff}}(x) = m_{\text{Cas}}(x), \quad m_{\text{ff}}(x) = m_{\text{Cas}}(x).$$

After restoring units, the corresponding energy is $m_{\text{Cas}}c^2$.

Proof. The D_x -term is quadratic in Δ and vanishes in the zero-transfer limit. By theorem 9.5, the numerator tends to $2(\bar{p}^0)^2$; division by $2\bar{p}^0$ gives $\bar{p}^0 = m_{\text{Cas}}$. $\square$

9.2 Improvement and completed-composite closure

Proposition 9.7 (Zero-transfer improvement invariance). *Let*

$$\hat{\mathbf{T}}_x^{\nu\mu} = \hat{\mathbf{T}}_x^{\mu\nu} + \partial_\rho \hat{B}_x^{\rho\mu\nu}$$

with a regular charge-neutral improvement in the fixed boundary class. Then the difference of the matrix elements is proportional to Δ_ρ and cannot change $A_x(0)$ or m_{ff} .

Proof. Translation covariance gives

$$\langle p' | \partial_\rho \hat{B}_x^{\rho\mu\nu}(0) | p \rangle = i\Delta_\rho \langle p' | \hat{B}_x^{\rho\mu\nu}(0) | p \rangle.$$

Regularity at zero transfer makes this term vanish there. Equivalently, the whole-space charge is unchanged by lemma 6.6. $\square$

Theorem 9.8 (Stable completed-composite normalization). *Let a stable completed composite define an isolated spin-zero one-particle sector of the full theory. Use the operator/source-attached total tensor reconstructed from the complete normalized-ray accepted Module A source-presentation family with coherent representatives and homogeneous extension and containing every certified constituent, binding,*

mediator, interaction, internal-stress, restoration, transport, baseline, and boundary contribution exactly once. Then

$$A_{\mathcal{B}}(0) = 1, \quad m_{\text{ff}}(\mathcal{B}) = m_{\text{Cas}}(\mathcal{B}).$$

No additivity of bare constituent masses is assumed.

Proof. The integrated attached complete tensor is the total translation generator by theorem 7.3 and proposition 6.3. The proof of theorem 9.5 therefore applies to the composite spectral sector without resolving the tensor into constituent pieces. Binding and stress data can change $D_{\mathcal{B}}(t)$, higher multipoles, and nonzero-transfer dependence, but not the zero-transfer translation normalization. $\square$

A partial constituent tensor, unbalanced internal-exchange ledger, nonconserved improvement, singular zero-transfer limit, or selected-state expectation used in place of the generalized-state matrix element does not satisfy this theorem.

10 Complete composite form-factor attachment

Definition 10.1 (Stable-sector attached form-factor family). For a matching object, the transition functional

$$h_{\mu\nu} \longmapsto \frac{1}{2} \int d^4x h_{\mu\nu}(x) \langle p' | \hat{\mathbf{T}}_x^{\mu\nu}(x) | p \rangle$$

is the rigged-space continuation of the operator/source attachment constructed in section 4. The certificate fixes the same upstream mass-bearing response, the same normalized-ray accepted Module A source-presentation family, the same coherent concrete representatives and homogeneous extension, the same attached operator $\hat{\mathbf{T}}_x$, all lawful improvements, and the generalized-state normalization data.

Proposition 10.2 (No selected-state expectation substitution). *The attachment in definition 10.1 is an off-diagonal transition distribution reconstructed from the complete norm-squared homogeneous extension of the normalized ray-state source family by polarization and continued to the stable rigged spectral domain. It is not obtained by choosing one normal state ω and replacing the form factor by $\omega(\hat{\mathbf{T}}_x^{\mu\nu})$. Therefore the theorem requires neither a normal sharp-momentum state nor a localized stationary wave packet.*

Proof. On the common dense domain, theorem 4.6 reconstructs every off-diagonal matrix element from the norm-squared homogeneous extension of the concrete accepted Module A normalized-ray source family. M3–M6 require the same attached form distribution to possess the tempered Poincaré spectral continuation used by generalized momentum states. This preserves the distinct mathematical types exposed by the first hostile review while retaining exact source ancestry. $\square$

Theorem 10.3 (Complete-composite zero-transfer closure). *For a stable spin-zero composite, every constituent, binding, field, interaction, stress, restoration, transport, baseline, and boundary contribution enters the total form factor through the operator/source-attached total tensor reconstructed from the complete accepted Module A normalized-ray composite source-presentation family and its homogeneous extension. The sum of all partial zero-transfer momentum fractions is one, and the total monopole is m_{Cas} .*

Proof. Accepted Module A supplies exact source-term coverage and balanced internal exchange for every normalized ray-state total source. Norm-squared extension and polarization are linear in the

source distributions, so the reconstructed attached operator is the corresponding exact registry sum as a form-valued distribution. Partial tensors need not be separately conserved and may carry scheme-dependent form factors. Their attached total is conserved and generates the full translations by theorem 7.3, so theorem 9.5 fixes the total coefficient to one. No bare constituent-mass additivity is used. $\square$

Part III

Active Metric-Source Readout

11 Linearized Einstein first response and the active readout

Restore c . In harmonic gauge about Minkowski space, with the module's fixed normalization,

$$\square \bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}, \quad \bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h.$$

For a stationary integrable source, the Green solution gives

$$h_{00}(\mathbf{r}) = \frac{\kappa}{4\pi} \int \frac{T^{00}(\mathbf{y}) + T^{ii}(\mathbf{y})}{|\mathbf{r} - \mathbf{y}|} d^3y.$$

The $1/r$ coefficient is therefore determined by the zero spatial Fourier component of the complete source combination.

Proposition 11.1 (Matrix-element compatibility of linear first response). *Let $\Theta_x^{\mu\nu}(p', p)$ be the operator/source-attached complete total-EMT elastic matrix element of section 9, and let $\mathcal{T}_{x,R}^{\mu\nu}$ be its regulated Breit transform. In the M8 linearized Einstein envelope, the Green operator acts linearly on tempered conserved sources. Hence applying the elastic generalized-state matrix element and then the lawful Breit smearing commutes with first-response propagation:*

$$\bar{h}_{x,R}^{\mu\nu} := -2\kappa G_{\square} * \mathcal{T}_{x,R}^{\mu\nu}$$

satisfies

$$\square \bar{h}_{x,R}^{\mu\nu} = -2\kappa \mathcal{T}_{x,R}^{\mu\nu}, \quad \partial_{\mu} \bar{h}_{x,R}^{\mu\nu} = 0$$

in the distributional sense, with the second identity understood in harmonic gauge. Equivalently, the same profile is obtained by solving the linearized equation in each elastic transfer channel and then applying the regulator transform.

Proof. The Breit source is tempered by M6–M7 and conserved by theorem 12.1. Convolution with the fixed linear Green operator is therefore a well-typed distributional operation. Linearity and continuity of convolution on the admitted tempered-source class permit the transfer integral and Green operator to be interchanged. Taking the divergence of the field equation and using source conservation preserves harmonic-gauge compatibility. No expectation value, probability density, or nonlinear metric equation is used. $\square$

The object $\bar{h}_{x,R}$ is a soft elastic *transition metric profile*: the first-response metric image of the attached total-EMT transition distribution. It is not asserted to be the expectation value of a classical metric in a localized normal quantum state. Module B extracts only its exact zero-transfer monopole, which is the frozen active-source readout.

Definition 11.2 (Active monopole coordinate). For a stationary source profile with

$$h_{00}(r) = \frac{A_g}{r} + O(r^{-2}),$$

define the active mass coordinate by $m_g = A_g/\beta_g$, where $\beta_g > 0$ is one fixed projection-wide source calibration. The value of β_g is not generated by Module B.

Remark 11.3. The old release applied this formula to a selected-state expectation source and then identified its energy with the sector Casimir. The corrected release instead constructs a stationary Breit source whose zero mode is fixed directly by the attached total-EMT Ward normalization.

12 Stationary Breit source and exact Laue closure

Work first in natural units. In the Breit frame set

$$p = (E_{\Delta/2}, -\Delta/2), \quad p' = (E_{\Delta/2}, \Delta/2), \quad \Delta^0 = 0.$$

Let $R \in \mathcal{S}(\mathbb{R}^3)$ be even with $R(0) = 1$. Define the regulated complete Breit profile

$$\mathcal{T}_{x,R}^{\mu\nu}(\mathbf{r}) := \int \frac{d^3\Delta}{(2\pi)^3 2E_{\Delta/2}} e^{-i\Delta \cdot \mathbf{r}} R(\Delta) \left\langle \frac{\Delta}{2} \left| \hat{\mathbf{T}}_x^{\mu\nu}(0) \right| -\frac{\Delta}{2} \right\rangle.$$

Theorem 12.1 (Breit-profile stationarity and conservation). *The profile is time independent and satisfies*

$$\partial_i \mathcal{T}_{x,R}^{i\nu} = 0$$

in the distributional sense.

Proof. The Breit transfer has $\Delta^0 = 0$, so the phase is independent of time. Conservation of the attached total-EMT matrix element gives $\Delta_\mu \langle p' | \hat{\mathbf{T}}_x^{\mu\nu} | p \rangle = 0$. Multiplication by the regulator and Fourier transformation turn this into the displayed spatial divergence identity. $\square$

Theorem 12.2 (Exact integrated energy and Laue stress). *The complete Breit profile satisfies*

$$\int d^3r \mathcal{T}_{x,R}^{00}(\mathbf{r}) = m_{\text{Cas}}, \quad \int d^3r \mathcal{T}_{x,R}^{ij}(\mathbf{r}) = 0.$$

After restoring units, the first identity is $\int T^{00} d^3r = m_{\text{Cas}} c^2$.

Proof. Spatial integration evaluates the Fourier transform at $\Delta = 0$. The regulator contributes $R(0) = 1$. In the rest limit, $E_0 = m_{\text{Cas}}$, $\bar{P}^0 = m_{\text{Cas}}$, $\bar{P}^i = 0$, and the D -tensor vanishes. Dividing the matrix element by the Breit normalization $2E_0$ gives

$$\frac{2(m_{\text{Cas}})^2 A(0)}{2m_{\text{Cas}}} = m_{\text{Cas}}$$

for the energy integral and zero for every spatial-stress integral. Use theorem 9.5. $\square$

Corollary 12.3 (Regulator independence of the monopole). *Any two lawful even Schwartz regulators with value one at the origin give the same integrated energy and Laue stress. Their differences affect only finite-size or short-distance profile data.*

The Breit profile is a stationary transition-source distribution, not the probability density of a normal localized one-particle state. It is precisely the device that permits simultaneous exact mass-shell normalization and stationary source extraction without the sharp-momentum/localization contradiction identified by the hostile referee.

13 Active soft-gravitational monopole from the total form factor

Theorem 13.1 (Active soft transition-metric monopole). *For the complete Breit source in the fixed first-response Einstein envelope,*

$$h_{00}(r) = \frac{\kappa c^2}{4\pi r} m_{\text{Cas}} + O(r^{-2}).$$

The $1/r$ coefficient is independent of $D_x(0)$, lawful improvement representative, internal composition, and regulator profile.

Proof. Apply proposition 11.1 and the stationary Green formula of section 11. By theorem 12.2,

$$\int (T^{00} + T^{ii}) d^3x = m_{\text{Cas}} c^2.$$

The coefficient of $1/r$ is therefore the displayed value. The D -term is proportional to two powers of momentum transfer and has vanishing zero mode; it can change contact, finite-size, or tidal terms, but not the monopole. Improvements also vanish at zero transfer. $\square$

Corollary 13.2 (Active scalar readout). *With the fixed source calibration,*

$$m_g(x) = \frac{\kappa c^2}{4\pi\beta_g} m_{\text{Cas}}(x).$$

Remark 13.3. The theorem proves an exact soft-source coefficient of the operator/source-attached complete total EMT. It does not identify m_{Cas} with $\omega(H)/c^2$ for a generic wave packet.

14 The active source line and exclusion of an object-dependent soft factor

Definition 14.1 (Active source line). For each matching object, $\mathcal{L}_{D,x}^g$ is the positive line generated by the zero-transfer metric monopole coefficient. The readout map is

$$\phi_g : \mathcal{L}_{D,x}^P \longrightarrow \mathcal{L}_{D,x}^g, \quad q \longmapsto \frac{\kappa c^2}{4\pi} q.$$

Theorem 14.2 (No independent attached total-EMT soft gravitational factor). *Inside the frozen attached total-EMT, universal-coupling envelope, an object-dependent factor γ_x in*

$$h_{00}(r) = \frac{\kappa c^2}{4\pi r} \gamma_x m_{\text{Cas}} + O(r^{-2})$$

requires $\gamma_x = 1$. A nonunit factor changes the total translation Ward normalization, source coupling, boundary condition, or calibration and therefore leaves the matching target.

Proof. The coefficient multiplying the total one-particle EMT monopole is $A_x(0)$. The global charge theorem forces $A_x(0) = 1$ for every stable object. A different value contradicts the translation charge, while a different metric coupling or calibration is an anchor change rather than a species coordinate in the fixed envelope. $\square$

Part IV

Poincare-to-Bargmann Inertial Readout

15 The explicit massive scalar Wigner representation

Fix $m = m_{\text{Cas}} > 0$ and

$$E_c(\mathbf{p}) = \sqrt{m^2 c^4 + c^2 |\mathbf{p}|^2}, \quad w_c(\mathbf{p}) = \frac{m c^2}{E_c(\mathbf{p})}.$$

Let

$$\mathcal{H}_c = L^2(\mathbb{R}^3, w_c d^3 p).$$

On $C_c^\infty(\mathbb{R}^3)$, define

$$\begin{aligned} (P_i^{(c)} \psi)(\mathbf{p}) &= p_i \psi(\mathbf{p}), & (H_c \psi)(\mathbf{p}) &= E_c(\mathbf{p}) \psi(\mathbf{p}), \\ (J_i^{(c)} \psi)(\mathbf{p}) &= -i\hbar \epsilon_{ijk} p_j \partial_{p_k} \psi(\mathbf{p}), & (K_i^{(c)} \psi)(\mathbf{p}) &= i\hbar \frac{E_c(\mathbf{p})}{c^2} \partial_{p_i} \psi(\mathbf{p}). \end{aligned}$$

Lemma 15.1 (Symmetry and invariant measure). *These operators are symmetric on C_c^∞ . The boost vector field is divergence free with respect to $w_c d^3 p$.*

Proof. For the boost field $v_i = E_c c^{-2} \partial_{p_i}$,

$$w_c v_i = \frac{m c^2}{E_c} \frac{E_c}{c^2} \partial_{p_i} = m \partial_{p_i},$$

whose weighted divergence vanishes. Integration by parts gives symmetry. The remaining operators are standard multiplication or rotation generators on the invariant measure. $\square$

Theorem 15.2 (Poincaré commutators). *On the common smooth core,*

$$[K_i^{(c)}, P_j^{(c)}] = i\hbar \delta_{ij} \frac{H_c}{c^2}, \quad [K_i^{(c)}, H_c] = i\hbar P_i^{(c)},$$

$$[K_i^{(c)}, K_j^{(c)}] = -\frac{i\hbar}{c^2} \epsilon_{ijk} J_k^{(c)},$$

with the usual rotation and translation relations. Their closures are the Stone generators of the massive scalar Wigner representation and have Casimir m .

Proof. The commutators follow by direct differentiation using $\partial_{p_i} E_c = c^2 p_i / E_c$. Exponentiating the complete boost and rotation flows gives the standard spin-zero induced representation [9]. Hence the closures are self-adjoint Stone generators. The Gårding domain is a common invariant core; the displayed compactly supported smooth functions lie in it. Complete-flow essential self-adjointness is also covered by the standard first-order criterion [12]. $\square$

16 Constructed common-core Poincaré-to-Bargmann contraction

Definition 16.1 (Controlled scalar contraction certificate). For the stable scalar Wigner sector of mass $m = m_{\text{Cas}}$, the certificate consists of:

- C1.** Hilbert spaces $\mathcal{H}_c = L^2(\mathbb{R}^3, w_c d^3p)$ carrying the exact massive scalar Wigner generators;
- C2.** unitaries $V_c : \mathcal{H}_c \rightarrow L^2(\mathbb{R}^3, d^3p)$;
- C3.** the common invariant core $\mathcal{S}(\mathbb{R}^3)$;
- C4.** rest-energy subtraction $H_c \mapsto H_c - mc^2 I$;
- C5.** strong common-core convergence of the transformed Hamiltonian, momentum, and boost generators;
- C6.** convergence on the same core of every product entering the Poincaré commutators;
- C7.** essential self-adjointness of the limiting Galilei generators and preservation of the fixed mass label and zero-transfer Ward normalization as matching data.

Define the unitary

$$V_c : \mathcal{H}_c \longrightarrow L^2(\mathbb{R}^3, d^3p), \quad (V_c \psi)(\mathbf{p}) = w_c(\mathbf{p})^{1/2} \psi(\mathbf{p}).$$

The transformed rest-subtracted generators are

$$\tilde{H}_c = V_c(H_c - mc^2 I)V_c^{-1} = E_c - mc^2, \quad \tilde{P}_i^{(c)} = p_i,$$

$$\tilde{K}_i^{(c)} = i\hbar \left(\frac{E_c}{c^2} \partial_{p_i} + \frac{p_i}{2E_c} \right).$$

Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^3)$.

Theorem 16.2 (Explicit graph and commutator convergence). *For every $\varphi \in \mathcal{S}$, as $c \rightarrow \infty$,*

$$\tilde{H}_c \varphi \longrightarrow \frac{|\mathbf{p}|^2}{2m} \varphi, \quad \tilde{P}_i^{(c)} \varphi \longrightarrow p_i \varphi,$$

$$\tilde{K}_i^{(c)} \varphi \longrightarrow i\hbar m \partial_{p_i} \varphi.$$

The products entering the Poincaré commutators converge on the same core, and the limiting operators are essentially self-adjoint there.

Proof. The inequalities

$$0 \leq E_c - mc^2 \leq \frac{|\mathbf{p}|^2}{2m}, \quad \frac{E_c}{c^2} - m \longrightarrow 0, \quad \frac{p_i}{2E_c} \longrightarrow 0$$

together with Schwartz decay give norm convergence by dominated convergence. The transformed commutators are exact:

$$[\widetilde{K}_i^{(c)}, p_j] = i\hbar\delta_{ij}\frac{E_c}{c^2}, \quad [\widetilde{K}_i^{(c)}, \widetilde{H}_c] = i\hbar p_i,$$

and the boost–boost commutator is $O(c^{-2})$ on $\mathcal{S}$. Thus the commutator products converge on the same core. The finite- c operators are unitarily equivalent to Wigner Stone generators. The limiting multiplication operators are essentially self-adjoint on $\mathcal{S}$, and $i\hbar m\partial_{p_i}$ is the self-adjoint translation generator in momentum space with $\mathcal{S}$ as a core. $\square$

Theorem 16.3 (Class-wide contraction inhabitation). *Every positive-mass spin-zero Wigner sector is unitarily equivalent to the construction above and therefore admits the frozen common-core contraction certificate. The Casimir label m is the central coefficient of the limiting representation. The independently proved matching datum $A(0) = 1$ remains fixed on the same object; it is not inferred from operator convergence.*

Proof. Wigner classification gives unitary equivalence to the standard induced representation. Transport the explicit core and generators through that equivalence. The mass is the fixed representation Casimir for every c . The total-EMT attachment and Ward-normalized form factor are transported separately by the matching certificate, so no localized source expectation is identified with the representation limit. $\square$

17 Bargmann central mass

Theorem 17.1 (Bargmann central-charge theorem). *The limiting generators satisfy*

$$[G_i, P_j] = i\hbar\delta_{ij} mI, \quad [G_i, H_{\text{nr}}] = i\hbar P_i, \quad [G_i, G_j] = 0.$$

Hence the Bargmann central mass is

$$M_B = mI = m_{\text{Cas}}I.$$

Proof. Take the limits in the exact commutators from theorem 16.2. The coefficient E_c/c^2 tends to m ; no new scalar is inserted after the limit. The remaining relations give the centrally extended Galilei algebra classified by Bargmann [10]. $\square$

Corollary 17.2 (Source identity survives the contraction). *The zero-transfer total-EMT monopole and the Bargmann central term are both labeled by the same Casimir m . This is a theorem about two representations of one stable-sector charge, not an equality of their codomains.*

18 Schrödinger inertial readout

Theorem 18.1 (Schrödinger coefficient matching). *In the minimal scalar nonrelativistic projection,*

$$H_{\text{kin}} = \frac{\mathbf{P}^2}{2m_{\text{Cas}}} = -\frac{\hbar^2}{2m_{\text{Cas}}} \Delta.$$

Therefore the inertial coordinate is

$$m_i(x) = m_{\text{Cas}}(x).$$

Proof. The explicit contraction gives the Bargmann central mass m_{Cas} and the free limiting Hamiltonian $\mathbf{P}^2/(2m_{\text{Cas}})$. The imported Schrödinger normal-form theorem fixes the local scalar position-space representation after the action scale $\hbar$ is admitted. The potential ledger does not change the kinetic mass coordinate. $\square$

Part V

Universal Proportionality and Naturality

19 One intrinsic charge line and two downstream readouts

The metric and Hamiltonian objects remain mathematically different. Their intrinsic scalar mass readouts are compared only because the active zero-transfer source and the contracted inertial representation have now been constructed from the same Poincaré Casimir/Noether charge.

Definition 19.1 (Readout maps). For $x \in \mathbf{Match}_D^{\text{mass}}$, define

$$\phi_i^x : \mathcal{L}_{D,x}^P \longrightarrow \mathcal{L}_{D,x}^i, \quad q \longmapsto qI,$$

and

$$\phi_g^x : \mathcal{L}_{D,x}^P \longrightarrow \mathcal{L}_{D,x}^g, \quad q \longmapsto \frac{\kappa c^2}{4\pi} q.$$

The first map is the Wigner/Bargmann contraction; the second is the exact zero-transfer attached total-EMT normalization followed by the Einstein first-response coupling. Both are positive line isomorphisms onto their images.

Theorem 19.2 (Common intrinsic charge factorization). *For every matching object, the inertial and*

active-source scalar branches factor as

$$\mathcal{L}_{D,x}^P$$

$$\text{arrow}[dl, \text{"}\phi_i^{x\prime\prime}\text{"}]$$

$$\text{arrow}[dr, \text{"}\phi_g^x\text{"}]$$

$$\mathcal{L}_{D,x}^i$$

$$\text{arrow}[d, \text{"}\chi_i\text{"}]$$

$$\mathcal{L}_{D,x}^g$$

$$\text{arrow}[d, \text{"}\chi_g\text{"}]$$

$$\mathbb{R}_{>0}$$

$$\mathbb{R}_{>0}.$$

The upper common object is m_{Cas} , equivalently the stable-sector Noether/Casimir charge. It is not a selected-state expectation energy, a localized-profile energy, or a formal identification of a Hamiltonian with a stress tensor.

Proof. The inertial branch is theorems 17.1 and 18.1. The active branch is theorems 9.5 and 13.1. Both begin with the translation generator identified with the integrated total stress-tensor charge on the full reference representation by theorem 7.3; the stable sector supplies the common positive Casimir coordinate. Their codomains and subsequent physical uses remain distinct. $\square$

Corollary 19.3 (Completion of the Module B portion of common descent). *On $\hat{Q}_{\text{mass}}^{\text{op}}$ restricted to the matching subtype, Hamiltonian inertia and intrinsic active metric-source response are faithful scalar representations of one upstream source-attached stable-sector charge.*

Remark 19.4. Accepted Module A supplies the full tensor-valued metric source, while the Schrödinger paper supplies the operator-valued Hamiltonian projection. Module B identifies only their common scalar zero-transfer/Casimir factor. Tensor anisotropy, spin, finite-size form factors, multipoles, and the state-dependent expectation energy are not collapsed into this scalar line.

20 One projection-wide proportionality factor

Theorem 20.1 (Universal inertial-active proportionality). *For every object $x \in \mathbf{Match}_D^{\text{mass}}$,*

$$m_g(x) = \alpha_D m_i(x), \quad \alpha_D := \frac{\kappa c^2}{4\pi\beta_g} > 0.$$

The same α_D applies to every stable scalar species and every completed stable scalar composite in the fixed matching envelope.

Proof. By theorem 18.1, $m_i = m_{\text{Cas}}$. By theorem 13.1, the intrinsic source amplitude is $\mathcal{A}_g = (\kappa c^2/4\pi)m_{\text{Cas}}$, and the source coordinate is $m_g = \mathcal{A}_g/\beta_g$. The coupling κ , Fourier/Green normalization, and calibration β_g are fixed projection anchors and carry no object index. The ratio is therefore one projection-wide scalar. $\square$

Corollary 20.2 (Completed-composite proportionality). *For a completed stable composite $\mathcal{B}$,*

$$m_g(\mathcal{B}) = \alpha_D m_i(\mathcal{B})$$

with both readouts determined by the intrinsic mass of the complete total translation/form-factor sector, including binding and internal response.

Proof. Apply theorems 9.8 and 17.1 to the stable composite sector. No constituent additivity assumption is used. $\square$

Theorem 20.3 (No unwitnessed species-indexed conversion). *Suppose a same-envelope model writes*

$$m_g^{(a)} = \alpha_a m_i^{(a)}.$$

Then either every $\alpha_a = \alpha_D$, or at least one difference is carried by a new admitted attached total-EMT form factor, source attachment, Einstein coupling, boundary, contraction, or calibration witness. In the latter case the model is a richer or different target. An unwitnessed varying family is inadmissible.

Proof. The matching theorem computes the active and inertial coordinates from $A_a(0) = 1$, the same Casimir mass, the fixed Einstein coupling, the canonical Wigner contraction, and the common calibration. If those data are unchanged, the ratio is α_D . A different ratio must change a load-bearing input. If no input changes, the extra factor performs unsupported selector work. $\square$

21 Naturality under lawful mass-bearing morphisms

Definition 21.1 (Charge, inertial, and source functors). Let

$$\mathcal{P}, \mathcal{I}, \mathcal{G} : \mathbf{Match}_D^{\text{mass}} \longrightarrow \mathbf{Line}_+$$

assign respectively the intrinsic Casimir/Noether charge line, inertial line, and active zero-transfer source line to each matching object, and the induced positive line isomorphism to each lawful matching morphism.

Theorem 21.2 (Readout naturality). *The families ϕ_i^x and ϕ_g^x are natural isomorphisms*

$$\phi_i : \mathcal{P} \Rightarrow \mathcal{I}, \quad \phi_g : \mathcal{P} \Rightarrow \mathcal{G}.$$

Consequently

$$\eta := \phi_g \circ \phi_i^{-1} : \mathcal{I} \Rightarrow \mathcal{G}$$

is a natural isomorphism. In fixed scalar coordinates, every component is multiplication by α_D .

Proof. A lawful morphism preserves the upstream response identity, complete normalized-ray accepted Module A source-presentation family with coherent representatives and homogeneous extension, operator/source attachment, attached total-EMT class, full translation action, charge-generation relation, stable Casimir sector, zero-transfer attached total-EMT normalization, canonical Wigner contraction, and projection-wide calibrations. The intrinsic charge therefore transports to the intrinsic charge, and both readout maps commute with the transported data. Composition and identity follow from the corresponding transport witnesses. The coordinate value is α_D by theorem 20.1. $\square$

Corollary 21.3 (Representation-independent ratio). *Basis changes, field redefinitions, lawful stress-tensor improvements, gauge representatives, generalized-state normalization conventions, and unitary equivalent Poincaré/Bargmann presentations leave m_g/m_i unchanged.*

Proposition 21.4 (Noninteracting monoidal compatibility). *For a disjoint noninteracting tensor product whose stable total mass is additive in the declared noninteracting envelope,*

$$m_i(x \otimes y) = m_i(x) + m_i(y), \quad m_g(x \otimes y) = m_g(x) + m_g(y).$$

For an interacting completed composite, the theorem applies to the total Casimir/form-factor charge rather than to a bare sum of constituent masses.

22 Flat calibration transport beyond the exact overlap

Theorem 22.1 (Unique flat transport extension). *Suppose every connected component of $\mathbf{Env}_D^{\text{mass}}$ meets $\mathbf{Match}_D^{\text{mass}}$ and relative calibration transport has trivial holonomy. Then the natural comparison established on $\mathbf{Match}_D^{\text{mass}}$ extends uniquely to that component.*

Proof. Choose a matching object x and a lawful path $f : x \rightarrow y$. Define

$$\eta_y := \tau_g(f) \eta_x \tau_i(f)^{-1}.$$

Flat relative transport makes the result path independent. Any other extension agreeing at x and natural along every arrow must equal this one. $\square$

Remark 22.2. Transport preserves a comparison already proved in the exact overlap. It does not create mass matching where the stable-sector, source, or contraction certificate is absent.

23 Common calibration and equality

Theorem 23.1 (Calibrated equality). *Under the common Einstein–Newton/Bargmann calibration*

$$\beta_g = \frac{\kappa c^2}{4\pi},$$

Module B gives

$$m_g(x) = m_i(x) = m_{\text{Cas}}(x)$$

throughout the calibration-connected fixed envelope.

Proof. The calibration sets $\alpha_D = 1$ in theorem 20.1; flat transport then applies theorem 22.1. $\square$

The equality is an equality of calibrated scalar readouts in the exact stable scalar envelope. It does not assert strong equivalence, generic curved-space global energy, or equality for massless, unstable, open, or radiating sectors.

Part VI

Inhabited Models and Bounded Audit Artifact

24 Inhabited and separating examples

The examples distinguish full matching objects from subtheorem witnesses. A table at the end of this section records exactly which M1–M10 coordinates each example inhabits.

24.1 Full inhabitant: free massive scalar field

Example 24.1 (Free Klein–Gordon operator/source matching object). Let x_m be the already licensed deformation-response role represented by a real free massive scalar field on Minkowski spacetime. Its two independent downstream witnesses are:

1. the compact metric-incidence response obtained from the first variation of

$$S[\phi, g] = -\frac{1}{2} \int \left(g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi + \frac{m^2 c^2}{\hbar^2} \phi^2 \right) d\mu_g;$$

2. the stable transport response carried by the positive-mass scalar Poincaré sector.

The first witness satisfies the accepted Module A attachment theorem and yields $\hat{x}_m \in \hat{Q}_{\text{src}}$; the second places $F(\hat{x}_m)$ in Q_{in} . Hence $\hat{x}_m \in \hat{Q}_{\text{mass}}$. This does not generate the upstream role from the field model; it constructs a faithful downstream occupant of the already established role.

Let $\mathfrak{D}_m$ be the finite-particle Schwartz core in the scalar Fock representation and let $\mathbb{P}_1(\mathfrak{D}_m)$ be its normalized ray space. Fix once and for all the Minkowski carrier, vacuum normal-ordering prescription, Hilbert stress-tensor improvement representative, regularity class, and boundary envelope. For every normalized ray $[\psi]$, the fixed state-evaluation protocol produces the prequotient accepted Module A

source presentation $\tilde{s}_{m, [\psi]}$ whose operational metric witness is

$$\mathcal{W}_{m, [\psi]}(h) = \frac{1}{2} \langle \psi, :T_m(h) : \psi \rangle, \quad \|\psi\| = 1.$$

Accepted Module A represents this witness by the concrete distribution $\mathbf{T}_{m, [\psi]}^{\text{ray}}$, and

$$[\mathbf{T}_{m, [\psi]}^{\text{ray}}]_{\simeq_{\text{law}}} = q_g(\mathfrak{s}_m^{\text{ray}}([\psi])).$$

Because the source presentation is formed before quotienting with one fixed normal-ordering and improvement representative, this is the coherent representative lift required by M2.

For an arbitrary vector $\varphi \in \mathfrak{D}_m$, define the homogeneous extension

$$\mathbf{T}_{m, \varphi}^{\sharp} = \begin{cases} \|\varphi\|^2 \mathbf{T}_{m, [\varphi/\|\varphi\|]}^{\text{ray}}, & \varphi \neq 0, \\ 0, & \varphi = 0. \end{cases}$$

Then, for every compact metric variation h ,

$$\langle \mathbf{T}_{m, \varphi}^{\sharp}, h \rangle = \langle \varphi, :T_m(h) : \varphi \rangle.$$

The physical source family is ray-invariant; the degree-two law belongs only to this auxiliary vector-weight extension. The finite-particle core is stable under addition, scalar multiplication, Poincaré transformations, and smeared field operations, so the homogeneous diagonal form satisfies the parallelogram identity and the required joint continuity. The polarization theorem therefore reconstructs the unique normal-ordered operator-valued distribution

$$\hat{\mathbf{T}}_m^{\mu\nu} =: T_m^{\mu\nu} :$$

whose normalized diagonal forms are exactly the concrete accepted Module A ray-state sources. Thus

$$(\hat{x}_m, \mathfrak{o}_m, \hat{\mathbf{T}}_m) \in \hat{\mathcal{Q}}_{\text{mass}}^{\text{op}}.$$

The Fock reference representation is Poincaré covariant and irreducible in the reference sense used here; the one-particle sector is the isolated spin-zero Wigner representation of mass m . The attached normal-ordered total stress tensor integrates to the translation generator on the full Fock representation.

On one-particle generalized states its matrix element is

$$\Theta^{\mu\nu}(p', p) = p'^{\mu} p^{\nu} + p'^{\nu} p^{\mu} - \eta^{\mu\nu} (p' \cdot p + m^2 c^2),$$

up to the common covariant normalization convention. At $p' = p$, the on-shell identity $p^2 = -m^2 c^2$ gives

$$\Theta^{\mu\nu}(p, p) = 2p^{\mu} p^{\nu},$$

so $A(0) = 1$. The canonical Wigner generators are exactly those of section 15; theorem 16.2 supplies C1–C7 and the Bargmann mass m . The zero-transfer Einstein source readout and the Schrödinger kinetic coefficient therefore factor through the same attached charge.

Proposition 24.2 (The free scalar object inhabits M1–M10). *The object of example 24.1 is a nontrivial*

object of $\mathbf{Match}_D^{\text{mass}}$.

Proof. M1 is the scalar deformation/inertial role and its accepted Module A source attachment. M2 is the explicit normalized-ray source-presentation family, coherent concrete lift, norm-squared homogeneous extension, and polarization reconstruction displayed above. M3–M5 are the free Fock reference representation and its isolated one-particle sector. M4 holds because the attached normal-ordered stress tensor generates translations on the full Fock space with vacuum baseline zero. M6 is the displayed regular matrix element of that same attached operator and theorem 9.5. M7 is the elastic Breit channel $q^0 = 0$. M8 is the admitted linearized Einstein coupling to the attached tensor. M9 is the explicit Wigner contraction. M10 is the fixed calibration line. No localized stationary normal state is invoked. $\square$

24.2 Separating source-only example

Example 24.3 (Accepted Module A source without operator presentation). A classical distributional source may possess a complete accepted Module A metric-response witness and therefore define an element of $\hat{Q}_{\text{mass}}$, while carrying no reference Hilbert space, no common invariant quantum domain, no normalized-ray Module A source-presentation family, and no coherent homogeneous quadratic extension. It fails M2 and does not define an element of $\hat{Q}_{\text{mass}}^{\text{op}}$. Such a source remains a valid Module A object but cannot be inserted into the operator form-factor/Bargmann theorem by adjoining an unrelated EMT.

24.3 Completed stable composite sector

Example 24.4 (Total-EMT composite form factor). Let an interacting theory possess an isolated stable scalar bound-state pole. Suppose accepted Module A certifies a complete normalized-ray total source-presentation family with one coherent prequotient representative family, and the M2 homogeneous-extension/polarization theorem reconstructs its attached operator total stress tensor, including constituent, binding, mediator, interaction, internal-stress, restoration, transport, and baseline terms. The bound-state generalized matrix element of that same attached operator has the scalar form-factor decomposition of definition 9.1. The integrated total charge normalization forces $A_{\mathcal{B}}(0) = 1$, while the nonzero-transfer functions encode finite-size and internal-stress structure. The module therefore compares the active and inertial intrinsic masses without replacing the composite mass by a sum of constituent labels.

24.4 Localized stationary Laue witness

Example 24.5 (Distributional point profile). Let a classical timelike worldline be $z^\mu(\tau) = (c\tau, 0, 0, 0)$ and

$$T^{\mu\nu}(x) = mc \int u^\mu u^\nu \delta^{(4)}(x - z(\tau)) d\tau.$$

In the rest frame,

$$T^{00} = mc^2 \delta^{(3)}(\mathbf{x}), \quad T^{0i} = T^{ij} = 0.$$

The profile is stationary and distributionally conserved, its Laue stress integral vanishes, and

$$h_{00} = \frac{\kappa mc^2}{4\pi r}.$$

This inhabits the localized-profile Laue and far-field subtheorems. It is not, by itself, a full quantum matching object because no full vacuum representation, operator form factor, or Wigner contraction has yet been attached.

24.5 Separating counterexample: a wave packet

Example 24.6 (Mean-rest wave packet is not the Casimir charge). Let ψ be a normalized nontrivial wave packet in the free scalar one-particle sector with symmetric momentum density. Then

$$\langle \mathbf{P} \rangle_\psi = 0, \quad \langle H \rangle_\psi > mc^2.$$

The sector Casimir and zero-transfer form-factor mass remain exactly m . This example proves that the repaired common charge cannot be the norm of the expectation four-momentum and explains why the form-factor theorem is load bearing.

24.6 Failure examples

Example 24.7 (Open radiating system). A source with nonzero outward energy flux may satisfy a local divergence equation with declared exchange, but its whole-space stationary profile charge and Laue balance need not exist. It is not an object of $\mathbf{Match}_D^{\text{mass}}$.

Example 24.8 (Massless or unstable sector). A stable massless helicity sector has no positive mass Casimir and does not contract to the massive Bargmann branch used here. A resonance does not define an isolated real-mass Wigner sector. Both are separately typed, not counterexamples.

Example 24.9 (Species-dependent multiplier). Take two objects with identical matching certificates except for a written factor $\alpha_1 \neq \alpha_2$ inserted between the same Casimir charge and source coordinate. Since no total-EMT form factor, source, boundary, contraction, or calibration witness changes, the difference is unsupported selector notation.

24.7 Example-status table

Example	Certificate coordinates	Status
Free massive scalar field	M1–M10	Full nontrivial $\mathbf{Match}_D^{\text{mass}}$ inhabitant.
Accepted Module A source without operator presentation	M1 only; M2 fails	Separating source-only object in $\hat{Q}_{\text{mass}} \setminus \hat{Q}_{\text{mass}}^{\text{op}}$; not eligible for form-factor/Bargmann matching.
Stable scalar bound-state sector	M1–M8, with M9–M10 conditional on the declared representation and calibration certificates	Class-wide completed-composite illustration; a concrete theory must supply the stable pole and matching data.

Example	Certificate coordinates	Status
Distributional point profile	Source/profile and Einstein first-response coordinates only	Laue/far-field subtheorem witness, not a quantum matching object.
Mean-rest wave packet	Stable sector and inertial representation only	Separating counterexample to expectation/Casimir identification; not the active matching source.

25 Bounded machine-readable certificate checker

The project contains a small Python checker for the finite rational coordinates of the Module B certificate. Its purpose is reproducible audit bookkeeping, not analytic proof.

The checker validates:

- the exact accepted Module A manuscript and positive-review hashes;
- module identity and corrected release version;
- presence of the finite certificate flags;
- one common source-ancestry identifier across the normalized-ray source family, concrete prequotient representative family, homogeneous vector extension, and attached operator;
- positivity of the displayed Casimir mass;
- the displayed zero-transfer normalization $A(0) = 1$;
- equality of the displayed intrinsic form-factor and Casimir masses;
- the active amplitude, Bargmann mass, Schrödinger mass, and calibrated active mass;
- one common displayed proportionality factor across all objects;
- exact completed-source term coverage and nonduplication;
- a vanishing displayed Laue stress trace only when a localized-profile subcertificate is declared applicable;
- deterministic certificate hashing.

It deliberately contains no field equating a selected-state expectation energy with the Casimir mass.

The checker does not verify Module A attachment, formation of a physical source family on normalized rays, existence of the concrete prequotient representative family, the norm-squared homogeneous extension, its quadratic identities, polarization continuity, operator-valued distribution existence, charge generation, irreducibility, generalized-state normalization, form-factor regularity, the Ward identity, PDE existence, Laue hypotheses, Wigner generator self-adjointness, graph convergence, commutator convergence, or physical adequacy. Boolean flags naming those coordinates are inputs to the finite audit, not evidence for them.

The documented command is

```
PYTHONPATH=software python3 -m unittest discover -s software/tests -v
python3 software/certificate.py ledgers/example_matching_certificate.json
```

The release test suite includes failure cases for a missing normalized-ray flag, a mismatched concrete representative-lift identifier, a mismatched homogeneous-extension identifier, an unrelated operator identifier, a wrong Module A hash, nonunit $A(0)$, mismatched intrinsic source mass, nonzero Laue trace when applicable, incorrect Bargmann mass, incomplete composite terms, and an object-dependent proportionality factor.

Part VII

Endpoint, Handoff, and Adversarial Closure

26 Main Module B theorems

Theorem 26.1 (Common Mass-Charge Matching Theorem). *Let $\hat{x} \in \mathbf{Match}_D^{\text{mass}}$. Then:*

1. *the complete normalized-ray accepted Module A source-presentation family, its coherent concrete prequotient representative family, and its norm-squared homogeneous vector extension reconstruct one unique operator-valued total stress distribution $\hat{\mathbf{T}}_x$, and every later charge and form factor is taken from that operator;*
2. *the attached operator has an integrated translation-charge system on the full reference representation whose diagonal forms are exactly the accepted Module A source charges;*
3. *that charge system equals the Stone translation generators after the fixed vacuum/baseline normalization;*
4. *the isolated stable scalar sector has one positive Casimir mass m_{Cas} ;*
5. *the attached total-stress matrix element satisfies the exact zero-transfer Ward normalization $A_x(0) = 1$, so its intrinsic active source mass is $m_{\text{ff}} = m_{\text{Cas}}$;*
6. *the explicit common-core Wigner contraction has Bargmann central mass $m_{\text{Cas}}I$;*
7. *the minimal scalar Schrödinger kinetic coefficient is $-\hbar^2/(2m_{\text{Cas}})$.*

Thus the intrinsic active metric-source and inertial Hamiltonian scalar readouts factor through one common Poincaré–Noether/Casimir charge while retaining distinct codomains and exact common ancestry in the upstream response object.

Proof. Item 1 is theorem 4.6 and definition 4.7; item 2 is proposition 6.4; item 3 is theorem 7.3; item 4 is definition 8.1; item 5 is proposition 9.2, theorems 9.5 and 13.1, and corollary 9.6; items 6–7 are theorems 16.2, 17.1 and 18.1. The common factorization is theorem 19.2. $\square$

Theorem 26.2 (Universal Source Naturality and Proportionality Theorem). *On the calibration-connected fixed envelope $\mathbf{Env}_D^{\text{mass}}$, the intrinsic active and inertial mass coordinates satisfy*

$$m_g^{(a)} = \alpha_D m_i^{(a)}$$

with one projection-wide factor α_D . The relation is natural under lawful mass-bearing morphisms and includes completed stable scalar composites. A varying species-indexed factor survives only as independently witnessed richer tensor/anchor content or target change.

Proof. The exact overlap theorem is theorems 20.1 and 21.2; the transport extension is theorem 22.1; exclusion of unwitnessed varying factors is theorem 20.3. $\square$

Corollary 26.3 (Calibrated equality). *Under the common Einstein–Newton/Bargmann calibration,*

$$m_g^{(a)} = m_i^{(a)} = m_{\text{Cas}}^{(a)}$$

throughout the fixed envelope.

Proof. Apply theorem 23.1. $\square$

Theorem 26.4 (Profile/Casimir nonconflation theorem). *For a generic normal state or localized profile, neither*

$$\frac{1}{c} \sqrt{-\eta_{\mu\nu} p_\omega^\mu p_\omega^\nu}$$

nor M_{src} is asserted to equal m_{Cas} . The common-charge theorem concerns the intrinsic zero-transfer attached total-EMT source mass and the contracted inertial representation. Equality with a particular profile energy requires an additional sharp intrinsic rest-charge matching witness.

Proof. This is the combined content of definition 2.6, proposition 2.7, and corollary 9.6. The localized Breit source used in sections 12 and 13 is a transition distribution, not a generic normal-state expectation profile. $\square$

27 Handoff to the modular capstone repair

Accepted Module A constructs the metric-source map and finite completed-source closure. Module B adds the exact stable scalar mass bridge:

$$\hat{Q}_{\text{mass}}^{\text{op}} \xrightarrow{\text{OpAttach}_g} \hat{\mathbf{T}}_x \longrightarrow m_{\text{Cas}} \longrightarrow \begin{cases} A_x(0) = 1 \longrightarrow \mathcal{A}_g, \\ \text{Wigner contraction} \longrightarrow M_B \longrightarrow -\hbar^2 \Delta / (2m_{\text{Cas}}). \end{cases}$$

The capstone may therefore import the following theorem object once Module B receives a positive independent re-review:

$$m_g = \alpha_D m_i$$

in the exact source-attached stable scalar matching and calibration-transport envelope.

The capstone must not promote Module B beyond its scope. In particular, Module B does not supply:

- the upstream nonidentity continuation required by Module C;

- spin, multipole, unstable-sector, radiating, or strong-equivalence closure;
- a generic curved-spacetime global energy;
- Module D fibre exhaustion or the final assignment orbit;
- numerical values of particle masses or of gravitational couplings.

Proposition 27.1 (Safe import interface). *A later manuscript may use Module B only after fixing:*

1. *the accepted Module A source attachment, the normalized-ray source-presentation family formed before quotienting, its coherent concrete representatives, and its homogeneous vector extension;*
2. *the polarization-reconstructed operator/source attachment on a common invariant domain;*
3. *the isolated stable scalar Poincaré sector;*
4. *the full-representation charge-generation package for the attached operator;*
5. *the regular zero-transfer form factor of that same attached operator;*
6. *the Einstein first-response and canonical Wigner contraction envelopes;*
7. *the common calibration and lawful transport certificate.*

Use outside those conditions is a new theorem target.

28 Adversarial closure and failure taxonomy

The first hostile-referee cycle identified the expectation/Casimir type error. The bounded second cycle confirmed that repair but found a further incidence gap: the accepted Module A source and the operator total EMT remained independent certificate coordinates. The bounded third cycle accepted the polarization construction conditionally but found two earlier source-family type errors: a physical projective-state map was simultaneously required to be degree-two homogeneous, and a lawful source-equivalence class was used as a concrete distribution. This release treats all three findings as governing adversarial constraints rather than wording issues.

28.1 Attacks closed by the repaired theorem chain

Attack	Exact closure
Projective state invariance conflicts with degree-two homogeneity.	Closed by definitions 4.1 and 4.2 and proposition 4.3. The physical source family is defined on normalized rays. Degree-two homogeneity belongs only to the auxiliary norm-squared vector-weight extension used for polarization.
A source quotient class is not a concrete distribution.	Closed before polarization: the normalized ray family takes values in accepted Module A source presentations prior to quotienting, and $\mathbb{T}_A$ constructs one concrete distribution on the fixed representative envelope. Its lawful class is then q_g of the derived quotient family. No state-dependent post-quotient representative choice occurs.

Attack	Exact closure
The accepted Module A source and the operator EMT are unrelated.	Closed by theorem 4.6 and definition 4.7. The operator tensor is uniquely reconstructed from the homogeneous extension of the concrete normalized-ray Module A source family; M4 and M6 use that same operator, and an unrelated current fails attachment.
The expectation four-momentum is not the Casimir.	Agreed and built into the type lock. The common active charge is obtained from the zero-transfer total-EMT form factor, not from $-\omega(P)^2$.
A sharp rest eigenstate is not a localized stationary normal state.	Agreed. Generalized momentum states are used only for spectral/form-factor normalization. Localized profile energy is separately typed.
A Breit transition source is not a normal-state classical source.	Agreed. It is typed as a conserved tempered transition distribution. Proposition 11.1 proves that linearized Einstein propagation commutes with elastic matrix-element extraction and lawful Breit smearing; only the soft transition monopole is reported.
Schur's lemma was used inside the wrong sector.	Charge/generator equality is proved on the full irreducible vacuum/reference representation and only then restricted to the stable sector.
The matching category might be empty.	The free massive scalar field, with its normalized-ray prequotient source family and explicit Wigner contraction, inhabits M1–M10.
The contraction was only a list of assumptions.	The canonical scalar Wigner generators, common core, graph limits, commutator limits, and self-adjoint realizations are written and proved explicitly.
Binding and stress may alter gravitational mass.	They alter the total Casimir and nonzero-transfer form factors. The complete total-EMT Ward identity still gives $A_B(0) = 1$; no bare constituent additivity is used.
Stress-tensor improvements change local densities.	Charge-neutral regular improvements are proportional to momentum transfer and cannot change the zero-transfer normalization. The normalized-ray prequotient family also fixes one improvement representative before polarization.
Laue stress is missing.	The complete stationary localized profile satisfies a separately proved Laue theorem. It is not used to identify a generic state energy with the Casimir.
Species-dependent gravitational factors may be inserted.	In the fixed envelope the active coefficient is fixed by $A(0) = 1$, κ , and one calibration. A different factor changes a witness or is unsupported selector work.

28.2 Routes that remain outside the endpoint

The following do not refute the module because they change or leave the target:

- massless, unstable, resonant, or confined-without-stable-particle sectors;

- nonzero radiative flux or an open source ledger;
- generic curved spacetimes without the exact matching and transport certificates;
- spin-dependent or multipole-dependent nonzero-transfer form factors;
- nonminimal richer-scope couplings that alter the total-EMT source certificate;
- self-force, strong-field backreaction, or strong-equivalence claims;
- a different Galilei projective cocycle or an unproved contraction;
- a numerical calibration loop with nontrivial relative holonomy.

28.3 Named failure points for bounded re-review

A bounded re-review should test, in order:

1. formation of the physical source family on normalized rays rather than arbitrary vectors;
2. construction of the prequotient concrete source family on one fixed representative envelope;
3. the well-defined norm-squared homogeneous extension and absence of a projective/homogeneity conflict;
4. complex-polarization reconstruction and uniqueness of the attached operator;
5. inheritance of support, naturality, Ward, boundary, improvement, and charge forms;
6. full-representation charge/generator equality;
7. the scalar form-factor decomposition, $A(0) = 1$, and improvement invariance;
8. the explicit Wigner generator formulas and common-core limits;
9. the active/inertial factorization and fixed calibration;
10. the repaired free scalar M1–M10 inhabitant and finite checker;
11. regressions in the accepted Breit, Laue, contraction, transport, and software subtheorems.

29 Conclusion

The repaired Module B closes the common scalar mass bridge without reversing the kernel dependency order and without equating mathematical objects that merely share the word “mass.” The upstream physical-interior and deformation-response roles remain prior. Accepted Module A supplies the metric-source attachment. Module B then forms a complete normalized-ray family of concrete prequotient source presentations on one fixed representative envelope and constructs its norm-squared homogeneous vector extension. On the operator-presentable subtype, the polarization theorem reconstructs one unique operator-valued total stress tensor whose normalized diagonal forms are exactly those accepted ray-state sources. The full Poincaré reference representation then supplies the comparison translation generator, and the integrated charge of that attached operator tensor is proved equal to the generator before any stable-sector restriction is made.

The stable sector contributes one positive Casimir mass m_{Cas} . A generic state expectation p_ω^μ and a localized-profile energy $M_{\text{src}}c^2$ remain separately typed and need not equal the Casimir. The intrinsic active source is instead fixed by the zero-transfer matrix element of the same operator/source-attached total stress tensor. Charge normalization gives

$$A_x(0) = 1, \quad m_{\text{ff}} = m_{\text{Cas}}.$$

This theorem includes stable completed composites because the attached total stress tensor and translation generator already contain binding, field, interaction, internal-stress, restoration, transport, and baseline contributions. Nonzero-transfer structure can retain finite-size and stress information without changing the intrinsic monopole normalization.

The inertial branch is constructed independently. The canonical positive-mass scalar Wigner representation admits a common-core contraction with

$$H_c - mc^2 \rightarrow \frac{\mathbf{p}^2}{2m}, \quad K_i^{(c)} \rightarrow i\hbar m \partial_{p_i},$$

and hence

$$[G_i, P_j] = i\hbar \delta_{ij} m I.$$

With $m = m_{\text{Cas}}$, the Bargmann mass and Schrödinger coefficient are the same intrinsic charge. The active and inertial readouts therefore factor through one line and satisfy

$$m_g = \alpha_D m_i.$$

One common Einstein–Newton/Bargmann calibration gives

$$m_g = m_i = m_{\text{Cas}}.$$

Laue stress balance and the coordinate-space far-field formula remain valid for complete localized stationary profiles. They govern M_{src} and provide a conventional-facing check, but they are no longer used to commit the invalid expectation/Casimir substitution. The free massive scalar field provides a nontrivial full matching inhabitant; a mean-rest wave packet provides the separating counterexample that enforces the repaired type discipline.

In compressed form,

$$\begin{aligned} & \text{accepted Module A normalized-ray source-presentation family} \implies \text{OpAttach}_g \implies \hat{\mathbf{T}}_x \\ & \implies \hat{Q}_x^\mu = \hat{P}^\mu \implies m_{\text{Cas}}, \\ & m_{\text{Cas}} \implies \begin{cases} A(0) = 1 \implies m_g, \\ \text{Wigner contraction} \implies m_i. \end{cases} \end{aligned}$$

followed by naturality and common calibration. No claim is made for a generic curved-spacetime global energy, strong equivalence, self-force, massless or unstable sectors, Module C continuation, Module D fibre uniqueness, or numerical mass prediction.

A Formal predicates and exact module interface

A.1 Mass-bearing overlap and normalized-ray operator-presentable subtype

For $\hat{x} = (x, \mathbf{a}_x, \mathbf{w}_x) \in \hat{Q}_{\text{src}}$, define

$$\text{MassBear}_D(\hat{x}) : \Longleftrightarrow F(\hat{x}) \in Q_{\text{in}}.$$

Then

$$\hat{Q}_{\text{mass}} = \{\hat{x} \in \hat{Q}_{\text{src}} : \text{MassBear}_D(\hat{x})\}.$$

Let

$$\text{SrcPres}_g^{\text{A}} = \{(p, \mathbf{a}_p, \mathbf{w}_p) : \text{SrcBear}_g(p, \mathbf{a}_p, \mathbf{w}_p)\}$$

be the accepted Module A source-bearing presentations before lawful quotienting. Let

$$\pi_{\text{A}} : \text{SrcPres}_g^{\text{A}} \rightarrow \hat{Q}_{\text{src}}, \quad \mathbb{T}_{\text{A}} : \text{SrcPres}_g^{\text{A}} \rightarrow \text{Src}_g^{\text{dist}}$$

be the Module A presentation quotient and concrete distributional representation maps.

Let $\mathfrak{D}_x \subset \mathcal{H}_{\text{ref}}$ be a dense invariant common domain and

$$\mathbb{P}_1(\mathfrak{D}_x) = \{\psi \in \mathfrak{D}_x : \|\psi\| = 1\} / U(1).$$

A normalized-ray source-presentation family over $\hat{x}$ is a map

$$\tilde{\mathfrak{s}}_x : \mathbb{P}_1(\mathfrak{D}_x) \rightarrow \text{SrcPres}_g^{\text{A}}$$

formed by one fixed state-evaluation protocol, preserving the same upstream mass-bearing response role and one common spacetime, boundary, regularity, normalization, and improvement envelope. Its quotient image and concrete source distribution are

$$\mathfrak{s}_x^{\text{ray}} = \pi_{\text{A}} \circ \tilde{\mathfrak{s}}_x, \quad \mathbf{T}_{x, [\psi]}^{\text{ray}} = \mathbb{T}_{\text{A}}(\tilde{\mathfrak{s}}_x([\psi])).$$

They satisfy

$$[\mathbf{T}_{x, [\psi]}^{\text{ray}}]_{\simeq_{\text{law}}} = q_g(\mathfrak{s}_x^{\text{ray}}([\psi])).$$

Define the homogeneous vector-weight extension

$$\mathbf{T}_{x, \psi}^{\#} = \begin{cases} \|\psi\|^2 \mathbf{T}_{x, [\psi/\|\psi\|]}^{\text{ray}}, & \psi \neq 0, \\ 0, & \psi = 0, \end{cases}$$

and

$$q_{x, h}^{\#}(\psi) = \langle \mathbf{T}_{x, \psi}^{\#}, h \rangle.$$

The ray-family predicate is

$$\begin{aligned} \text{RaySrc}_g(\hat{x}, \tilde{\mathfrak{s}}_x) : \Longleftrightarrow & \text{OneUpstreamRole} \wedge \text{PrequotientForm} \wedge \text{FixedSourceSlice} \\ & \wedge \text{RayTotal} \wedge \text{RayCov} \wedge \text{RayComplete}. \end{aligned}$$

The homogeneous quadratic predicate is

$$\begin{aligned}
\text{QuadSrc}_g(\hat{x}, \mathfrak{o}_x) : \iff & \text{RaySrc}_g(\hat{x}, \tilde{\mathfrak{s}}_x) \\
& \wedge q_{x,h}^\#(\lambda\psi) = |\lambda|^2 q_{x,h}^\#(\psi) \\
& \wedge q_{x,h}^\#(\phi + \psi) + q_{x,h}^\#(\phi - \psi) = 2q_{x,h}^\#(\phi) + 2q_{x,h}^\#(\psi) \\
& \wedge \text{JointContinuous}(q_{x,h}^\#; \phi, \psi, h) \\
& \wedge \text{HomComplete}(\tilde{\mathfrak{s}}_x).
\end{aligned}$$

The degree-two law is derived from the norm-squared extension; it is not imposed on the physical projective state family.

With the fixed convention, let $\text{Pol}(q^\#)$ denote

$$\frac{1}{4} [q^\#(\phi + \psi) - q^\#(\phi - \psi) - iq^\#(\phi + i\psi) + iq^\#(\phi - i\psi)].$$

The operator/source attachment predicate is

$$\begin{aligned}
\text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x) : \iff & \text{QuadSrc}_g(\hat{x}, \mathfrak{o}_x) \\
& \wedge \forall h, \phi, \psi, \quad \langle \phi, \hat{\mathbf{T}}_x(h)\psi \rangle = \text{Pol}(q_{x,h}^\#)(\phi, \psi) \\
& \wedge \hat{\mathbf{T}}_x \in \mathcal{L}(\Gamma_c(S^2 T^* M), \mathcal{L}(\mathfrak{D}_x, \mathfrak{D}'_x)) \\
& \wedge \text{SupportNaturalWardBoundaryImprovementCompatible}(\hat{\mathbf{T}}_x).
\end{aligned}$$

The structured operator-presentable subtype is

$$\hat{Q}_{\text{mass}}^{\text{op}} = \{(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x) : \hat{x} \in \hat{Q}_{\text{mass}} \wedge \text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x)\} / \sim_{\text{op}}.$$

The equivalence $\sim_{\text{op}}$ transports the normalized-ray prequotient source family, its fixed concrete representatives, the homogeneous extension, the common core, and the reconstructed operator.

A.2 Matching certificate

Write

$$\text{MatchCert}_D(\hat{x}) := M_1 \wedge \cdots \wedge M_{10},$$

where:

$$\begin{aligned}
M_1 &:= \hat{x} \in \hat{Q}_{\text{mass}}, \\
M_2 &:= \text{OpAttach}_g(\hat{x}, \mathfrak{o}_x, \hat{\mathbf{T}}_x), \\
M_3 &:= \text{FullIrredPoincareReference}(\mathcal{H}_{\text{ref}}, \mathfrak{D}_x, U, \Omega), \\
M_4 &:= \text{GlobalCurrentChargeGeneration}(\hat{\mathbf{T}}_x, \hat{Q}_x, \hat{P}), \\
M_5 &:= \text{StableScalarMassSector}(E_{m_{\text{Cas}}}, m_{\text{Cas}}), \\
M_6 &:= \text{RegularElasticAttachedEMTFormFactor}(\hat{\mathbf{T}}_x, \Theta_x, A_x, D_x), \\
M_7 &:= \text{ElasticBreitChannel}(q^0 = 0, q \rightarrow 0) \wedge [\text{LocalizedProfileUsed} \Rightarrow \text{LaueCert}], \\
M_8 &:= \text{EinsteinFirstResponse}(\eta, h, \kappa), \\
M_9 &:= \text{CanonicalWignerContraction}(m_{\text{Cas}}, \mathcal{S}(\mathbb{R}^3)), \\
M_{10} &:= \text{CalibrationTransport}(\mathcal{L}_D^i, \mathcal{L}_D^g).
\end{aligned}$$

The object class is

$$\text{Obj}(\mathbf{Match}_D^{\text{mass}}) = \{\hat{x} : \text{MatchCert}_D(\hat{x})\}.$$

M4 and M6 are explicitly about the same $\hat{\mathbf{T}}_x$ reconstructed in M2.

A.3 Full-representation charge predicate

Let

$$\text{GlobalChargeMatch}(\hat{x}, \hat{P})$$

mean:

1. the attached operator tensor $\hat{\mathbf{T}}_x$ has integrated essentially self-adjoint charges on the common invariant core;
2. their diagonal forms equal the accepted Module A source charges;
3. the charges exponentiate to a strongly continuous unitary translation group;
4. that group and the imported $U(a)$ implement the same automorphisms of the full irreducible observable representation;
5. both preserve the invariant zero-charge baseline;
6. charge-neutral operator/Module A improvements preserve the group.

The theorem proves

$$\text{MatchCert}_D(\hat{x}) \implies \text{GlobalChargeMatch}(\hat{x}, \hat{P}) \wedge \overline{\hat{Q}_x^\mu} = \hat{P}^\mu.$$

A.4 Zero-transfer attached form-factor predicate

For covariantly normalized generalized states, define

$$\Theta_x^{\mu\nu}(p', p) = \langle p' | \hat{\mathbf{T}}_x^{\mu\nu}(0) | p \rangle.$$

The scalar matching predicate is

$$\Theta_x^{\mu\nu} = 2\bar{P}^\mu \bar{P}^\nu A_x(t) + \frac{1}{2}(\Delta^\mu \Delta^\nu - \eta^{\mu\nu} \Delta^2) D_x(t),$$

with A_x, D_x regular at $t = 0$. Charge normalization proves

$$A_x(0) = 1.$$

In natural units, the intrinsic source-mass predicate is

$$m_{\text{ff}}(x) := \lim_{\substack{\Delta \rightarrow 0 \\ \Delta^0 = 0}} \frac{\Theta_x^{00}(\bar{p} + \Delta/2, \bar{p} - \Delta/2)}{2\bar{p}^0} = m_{\text{Cas}}(x).$$

Restoring units, the corresponding zero-transfer source energy is $m_{\text{Cas}}c^2$.

A.5 Localized-profile predicate

If a stationary localized profile is separately invoked, define

$$M_{\text{src}} := \frac{1}{c^2} \int T^{00} d^3x.$$

Laue closure proves the profile far field

$$h_{00}(r) = \frac{\kappa c^2 M_{\text{src}}}{4\pi r} + O(r^{-2}).$$

No general predicate identifies M_{src} with m_{Cas} .

A.6 Contraction and universal factor predicates

The canonical contraction predicate is the conjunction of C1–C7 in definition 16.1, witnessed by the explicit Wigner generators. Its conclusion is

$$M_{\text{B}} = m_{\text{Cas}} I.$$

Let χ_i and χ_g be the fixed inertial and source scalar coordinates. Define

$$\alpha_D = \frac{\kappa c^2}{4\pi\beta_g}.$$

Then

$$\text{UniversalFactor}_D : \Longleftrightarrow \forall x \in \text{Obj}(\mathbf{Env}_D^{\text{mass}}), \quad m_g(x) = \alpha_D m_i(x).$$

Exact operator/source matching and flat calibration transport prove the predicate on each connected component.

B Laue balance, zero-transfer source, and far-field details

B.1 Distributional Laue argument

Let T^{ij} be a stationary distribution with compact spatial support and $\partial_i T^{ij} = 0$. Choose a cutoff χ_R equal to one on the support for large R . For the test function $x^k \chi_R$, distributional conservation gives

$$0 = \langle \partial_i T^{ij}, x^k \chi_R \rangle = -\langle T^{ij}, \delta_i^k \chi_R + x^k \partial_i \chi_R \rangle.$$

For sufficiently large R , the cutoff derivative is disjoint from the support, so

$$\langle T^{kj}, 1 \rangle = 0.$$

For noncompact support, the same proof applies when the boundary term tends to zero and the first moments exist.

B.2 Trace reversal and the localized active combination

With $\eta = \text{diag}(-1, 1, 1, 1)$,

$$h_{00} = \frac{1}{2}(\bar{h}_{00} + \bar{h}_{ii}).$$

The Green solution to $\nabla^2 \bar{h}_{\mu\nu} = -2\kappa T_{\mu\nu}$ is

$$\bar{h}_{\mu\nu}(\mathbf{x}) = \frac{\kappa}{2\pi} \int \frac{T_{\mu\nu}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d^3y.$$

Hence

$$h_{00} = \frac{\kappa}{4\pi} \int \frac{T^{00} + T^{ii}}{|\mathbf{x} - \mathbf{y}|} d^3y.$$

Laue closure removes the integrated stress term for a complete stationary profile.

B.3 Intrinsic zero-transfer channel

For a stable scalar sector, the intrinsic active source is instead read from

$$\Theta^{\mu\nu}(p', p) = 2P^\mu P^\nu A(t) + \frac{1}{2}(q^\mu q^\nu - \eta^{\mu\nu} q^2)D(t).$$

At $q = 0$, the stress term vanishes and charge normalization gives $A(0) = 1$. In the rest channel,

$$c \frac{\Theta^{00}(\bar{p}, \bar{p})}{2\bar{p}^0} = m_{\text{Cas}} c^2.$$

This intrinsic channel is independent of whether a normal localized stationary state exists.

B.4 Why the two channels are not conflated

A stationary profile coefficient measures $M_{\text{src}} = E_{\text{profile}}/c^2$. The intrinsic stable sector coefficient measures m_{Cas} . They agree only with an additional sharp intrinsic rest-charge matching witness. A generic wave packet has extra center-of-mass kinetic energy and is not such a witness.

B.5 Gauge and improvement boundary

The monopole coefficient is evaluated in the fixed asymptotically Minkowskian harmonic-gauge certificate. Residual gauge transformations preserving the falloff do not change the scalar coefficient. A regular improvement contributes a factor of q to the form factor and a surface term to the profile charge; both vanish in the fixed zero-transfer/no-flux class.

B.6 Einstein–Newton calibration

If $\kappa = 8\pi G_D/c^4$, the intrinsic amplitude is

$$\mathcal{A}_g = \frac{2G_D}{c^2} m_{\text{Cas}}.$$

Choosing $\beta_g = 2G_D/c^2$ makes the calibrated active coordinate equal to m_{Cas} .

C Explicit Wigner contraction and domain audit

C.1 Weighted Wigner space and unitary identification

For fixed $m > 0$, let

$$E_c(\mathbf{p}) = \sqrt{m^2 c^4 + c^2 |\mathbf{p}|^2}, \quad w_c(\mathbf{p}) = \frac{mc^2}{E_c(\mathbf{p})},$$

$$\mathcal{H}_c = L^2(\mathbb{R}^3, w_c d^3p), \quad (V_c \psi)(\mathbf{p}) = w_c(\mathbf{p})^{1/2} \psi(\mathbf{p}).$$

The maps V_c are unitary onto $\mathcal{H} = L^2(\mathbb{R}^3, d^3p)$. On the common core $\mathcal{S}(\mathbb{R}^3)$, the transformed rest-subtracted generators are

$$\tilde{H}_c = E_c - mc^2, \quad \tilde{P}_i = p_i,$$

$$\tilde{K}_i^{(c)} = i\hbar \left(\frac{E_c}{c^2} \partial_{p_i} + \frac{p_i}{2E_c} \right).$$

C.2 Common-core convergence

For every $\varphi \in \mathcal{S}$,

$$\left\| (E_c - mc^2)\varphi - \frac{|\mathbf{p}|^2}{2m}\varphi \right\| \rightarrow 0,$$

$$\left\| \tilde{K}_i^{(c)}\varphi - i\hbar m \partial_{p_i}\varphi \right\| \rightarrow 0.$$

The pointwise bounds

$$0 \leq E_c - mc^2 \leq \frac{|\mathbf{p}|^2}{2m}, \quad \frac{E_c}{c^2} \rightarrow m, \quad \frac{p_i}{2E_c} \rightarrow 0$$

and Schwartz decay give dominated convergence. This is the strong common-core graph convergence required by C5; no convergence of arbitrary domains is inferred.

C.3 Commutator products

On $\mathcal{S}$, the exact relations are

$$[\widetilde{K}_i^{(c)}, p_j] = i\hbar\delta_{ij}\frac{E_c}{c^2}, \quad [\widetilde{K}_i^{(c)}, \widetilde{H}_c] = i\hbar p_i,$$

$$[\widetilde{K}_i^{(c)}, \widetilde{K}_j^{(c)}] = -\frac{i\hbar}{c^2}\epsilon_{ijk}J_k.$$

The products have been evaluated before the limit, on one fixed core. Therefore

$$[G_i, P_j] = i\hbar\delta_{ij}mI, \quad [G_i, H_{\text{nr}}] = i\hbar P_i, \quad [G_i, G_j] = 0.$$

C.4 Essential self-adjointness

At finite c , the generators are the Stone generators of the complete massive Wigner flows. Their transformed versions are unitarily equivalent. The limiting Hamiltonian and momenta are real multiplication operators and $G_i = i\hbar m\partial_{p_i}$ generates momentum translations; $\mathcal{S}$ is a common core. This proves C7 in the explicit model.

C.5 Source identity and the scope of the contraction

The contraction acts on the stable representation. The source attachment and the total-EMT Ward normalization are independently proved matching data. The certificate requires that lawful Wigner identifications preserve the same upstream response class, accepted Module A source attachment, and form-factor normalization. No normal-state source density is transported through the contraction.

C.6 Schrödinger representation

The limiting Bargmann representation has central mass m . Translation invariance and $[G_i, H] = i\hbar P_i$ give

$$H = \frac{\mathbf{P}^2}{2m} + E_0.$$

The imported Schrödinger normal-form theorem supplies locality, scalarity, self-adjoint domain, and potential-ledger control.

C.7 Failure modes

The contraction leaves the frozen target when the sector is massless or unstable, the common core is absent, rest-energy subtraction is unavailable, commutator products fail to converge, the limit is open or nonunitary, another scaling produces a different kinematic contraction, or a new central coefficient is inserted after the limit.

D Certificate schema and machine-readable fields

The bounded JSON checker records only finite rational display data. It is not evidence for the analytic, distributional, or representation-theoretic premises.

Field	Meaning
module	Module identity, corrected version, accepted Module A hashes, and bounded-review provenance.
calibration	Global rational display values for the source prefactor, source calibration, and one common α_D .
objects	Finite example records for stable operator-presentable matching objects.
certificate_flags. operator_source_ attachment	Finite declaration that the object carries the analytic M2 attachment certificate. The checker does not prove that certificate.
source_attachment_id	Identifier for the accepted Module A normalized-ray source-presentation family used by the example.
ray_source_family_id	Identifier for the normalized physical ray-state source family.
source_representative lift_id	Identifier for the concrete prequotient source-presentation family on the fixed representative envelope.
homogeneous extension_id	Identifier for the norm-squared vector-weight extension used in polarization.
operator_attachment_id	Identifier for the polarization-reconstructed operator source. The finite checker requires all source-ancestry identifiers in this block to coincide.
object.id	Unique identifier.
casimir_mass	Positive stable-sector Casimir coordinate.
form_factor_A0	Displayed zero-transfer total-EMT normalization; required to equal one.
ff_mass	Intrinsic source mass extracted from the zero-transfer form factor.
active_ amplitude	First-response metric amplitude computed from the intrinsic source mass.
bargmann_mass	Central mass of the contracted representation.
schrodinger_ mass	Kinetic denominator mass coordinate.
active_mass	Source coordinate after the declared β_g calibration.
object.laue_applicable	Whether a separate localized stationary-profile subcertificate is declared.
laue_trace	Required to vanish only when laue_applicable is true.
terms	Exact completed-source term registry.

The checker first rejects a missing normalized-ray/operator flag or a mismatch among the finite ray-family, concrete-lift, homogeneous-extension, and operator-attachment ancestry identifiers. It then rejects a nonpositive Casimir mass, $A(0) \neq 1$, mismatch between intrinsic form-factor mass and Casimir mass, incorrect active amplitude, incorrect Bargmann or Schrödinger mass, object-dependent proportionality, invalid Laue display when applicable, and incomplete composite term coverage. It

does not verify the common dense domain, normalized-ray source formation, concrete prequotient lift, homogeneous extension, quadratic identities, polarization reconstruction, operator-valued distribution, or any analytic theorem. It also contains no field equating a normal-state expectation energy to the Casimir.

E Theorem-use and provenance ledger

Source	Exact use	Prohibited promotion
AASC kernel	Dependency order, no backflow, no second classifier.	No metric, mass, or charge generated at the boundary.
Realized Physical Interior	Particle, field, transport, mass-response, vacuum, time, and causal roles.	No constructive Lagrangian or completed dynamics imported.
Mass, Inertia, and Persistence Compression	Upstream deformation-response role μ_D and inertial restriction ι_D .	No numerical mass or equivalence result imported.
Accepted Module A	Source-bearing subtype, prequotient attachment-certified tensor distributions, Ward closure, and completed composite source.	No projective state family, operator EMT, Poincaré Casimir, Bargmann limit, or universal proportionality imported.
Module B polarization bridge	Forms a normalized-ray family of accepted Module A source presentations on one fixed prequotient envelope, takes its norm-squared homogeneous vector extension, and reconstructs one operator-valued distribution on a common dense invariant domain.	Does not promote every Module A source to operator-presentable status, treat non-normalized vectors as new physical states, choose source representatives after quotienting, or insert an independent current.
Spacetime Symmetry as Domain Anchor	Poincaré action and generators, stable massive sectors, current/charge typing, and improvement discipline.	A stress tensor or charge is not inferred from abstract covariance alone.
Pagels; Hudson–Schweitzer; Maynard	Spin-zero total-EMT form-factor basis and free-field benchmark.	The normalization $A(0) = 1$ is proved from the module’s global charge theorem, not imported.
Wigner; Bargmann; Inönü–Wigner	Massive spin-zero representation, Galilei central extension, and contraction comparison.	They do not replace the explicit common-core generator construction.
Einstein Dynamics	Minimal first-response Einstein equation and fixed coupling κ .	No generic curved-space global mass or strong-equivalence result imported.

Source	Exact use	Prohibited promotion
Schrödinger Dynamics	Minimal scalar Hamiltonian normal form.	No Poincaré-to-Bargmann bridge imported.
Laue literature	Conditions and interpretation of global stress integrals.	It does not bypass exact stationarity, conservation, zero-mode, and boundary certificates.

F Claim-status and nonclaim matrix

Claim	Status	Boundary
Operator/source attachment	Proved on $\hat{Q}_{\text{mass}}^{\text{op}}$	Requires a total quadratic normalized-ray source-presentation family on one common dense invariant domain; not every Module A source qualifies.
Profile source charge is conserved	Proved conditionally	Requires accepted Module A closure, integrability, and zero flux.
Full attached operator source charge equals translation generator	Proved	Uses exactly the polarization-reconstructed operator, the full irreducible reference representation, charge generation, exponentiation, and vacuum baseline.
Stable Casimir mass	Constructed from imported representation	Isolated positive-mass scalar sector only.
Expectation energy equals Casimir	Explicitly not claimed	False for generic wave packets.
Total-EMT zero-transfer normalization $A(0) = 1$	Proved	Regular elastic stable scalar form-factor channel.
Stable composite zero-transfer normalization	Proved conditionally	Requires isolated stable scalar composite sector and complete total tensor.
Laue stress balance	Proved conditionally	Regulated stationary conserved Breit transition profile.
Intrinsic active metric monopole	Proved	Zero-transfer matrix element of the same operator attached to the accepted Module A source family, plus linear first-response compatibility.

Claim	Status	Boundary
Localized profile far field	Proved conditionally	Separate positive/profile subclass; coefficient uses M_{src} , not automatically m_{Cas} .
Controlled Wigner contraction	Constructed and proved	Positive-mass scalar Wigner sector and common core.
Schrödinger coefficient matching	Proved	Minimal scalar nonrelativistic projection.
Universal $m_g = \alpha_D m_i$	Proved	Calibration-connected fixed scalar envelope.
Calibrated $m_g = m_i$	Proved after common calibration	No numerical particle mass or coupling derived.
Strong equivalence	Not claimed	Self-gravity, local experiments, all-order effects outside scope.
Generic curved global energy	Not claimed	No Poincaré/form-factor matching certificate.
Unique full certificate orbit	Not claimed	Later Module D.
Upstream GR/QM continuation	Not claimed	Later Module C.

G Standalone hostile-referee checklist

Before issuing a verdict, the reviewer should answer each question with the earliest failed theorem label if the answer is negative.

1. Is the kernel dependency order preserved without metric, Poincaré, Bargmann, form-factor, or mass backflow?
2. Is the source-bearing domain imported only through accepted Module A?
3. Is $\hat{Q}_{\text{mass}}$ the exact intersection of independently formed source and inertial subtypes?
4. Is $\hat{Q}_{\text{mass}}^{\text{op}}$ formed by a common dense domain and a total quadratic family of accepted Module A normalized-ray source presentations rather than by adjoining an operator current?
5. Does complex polarization reconstruct one unique $\hat{\mathbf{T}}_x$, and do M4 and M6 use exactly that operator?
6. Are m_{Cas} , p_ω , and M_{src} kept distinct?
7. Is the physical source family defined on normalized rays, formed at the Module A prequotient level, and represented concretely on one fixed source envelope before the norm-squared homogeneous extension is polarized?
8. Are support, naturality, Ward, boundary, improvement, and integrated charge identities first verified on the concrete normalized-ray sources, then extended homogeneously and transported to the operator by polarization?

9. Is charge/generator equality proved on the full irreducible reference representation before stable-sector restriction?
10. Are generalized momentum states used only distributionally, without being called localized normal sources?
11. Is the Breit profile explicitly typed as a transition distribution, and is its metric image justified by the linear first-response compatibility theorem rather than by a normal-state expectation?
12. Does the total-EMT matrix element have the correct scalar decomposition and regularity at zero transfer?
13. Does $A(0) = 1$ follow from the integrated charge, rather than being assumed?
14. Are stress-tensor improvements shown unable to change the zero-transfer normalization?
15. Does the stable-composite theorem use the complete total tensor rather than bare constituent masses?
16. Is the intrinsic active monopole derived from the zero-transfer source and fixed Einstein coupling?
17. Is Laue balance retained only for the localized stationary profile subclass?
18. Are the Wigner generators, common core, graph convergence, commutator convergence, and self-adjointness explicitly constructed?
19. Is the Bargmann central charge derived from the Casimir mass rather than inserted?
20. Is the Schrödinger coefficient matched on the correct scalar self-adjoint domain?
21. Is α_D projection-wide before calibration?
22. Is the free massive scalar field a genuine M1–M10 inhabitant?
23. Are strong equivalence, generic curved energy, and complete capstone closure excluded from the report?
24. Does the software remain bounded to finite algebraic certificate checks?

Verdict form

The report should issue separate verdicts for:

- source admission and operator/source attachment;
- full attached-source-charge/generator equality;
- zero-transfer total-EMT normalization and active monopole;
- localized-profile Laue/far-field subtheorem;
- explicit Bargmann contraction;
- universal proportionality/naturality;
- inhabited-example burden;

- software correspondence;
- overall Module B.

H Symbol index

Symbol	Meaning
Q_{resp}	Complete upstream deformation-response quotient.
Q_{in}	Response classes carrying a complete inertial witness.
$Q_{\text{resp}}^{\text{src}}$	Base source-bearing subtype constructed by accepted Module A.
$\hat{Q}_{\text{src}}$	Structured source attachment/witness quotient; domain of Module A q_g .
Q_{mass}	Exact base overlap $Q_{\text{in}} \cap Q_{\text{resp}}^{\text{src}}$.
$\hat{Q}_{\text{mass}}$	Structured mass-bearing refinement $F^{-1}(Q_{\text{mass}})$.
$\hat{Q}_{\text{mass}}^{\text{op}}$	Operator-presentable mass-bearing subtype with a total normalized-ray source-presentation family and polarization-reconstructed operator source.
$\text{SrcPres}_g^{\text{A}}$	Accepted Module A source-bearing presentations before lawful source quotienting.
$\mathbb{P}_1(\mathfrak{D}_x)$	Normalized projective state space: unit vectors in the common core modulo phase.
$\tilde{\mathfrak{s}}_x$	Fixed prequotient normalized-ray source-presentation family.
$\mathbf{T}_{x, [\psi]}^{\text{ray}}$	Concrete accepted Module A source distribution of a normalized physical ray.
$\mathbf{T}_{x, \psi}^{\#}$	Norm-squared homogeneous vector-weight extension used for polarization; not a physical-state map.
$\mathfrak{D}_x$	Common dense invariant graph-topology domain for the normalized-ray source-presentation family and attached operator forms.
$\mathfrak{D}'_x$	Continuous anti-dual of $\mathfrak{D}_x$.
OpAttach_g	Incidence relation identifying the unique operator-valued distribution reconstructed from the accepted Module A homogeneous extension of the normalized ray-state source family.
$\hat{\mathbf{T}}_x$	Polarization-reconstructed Hermitian operator-valued stress-energy distribution attached to object x .
$\text{Match}_D^{\text{mass}}$	Exact Poincaré–Noether–form-factor–Einstein–Bargmann matching category.
$\text{Env}_D^{\text{mass}}$	Calibration-connected lawful transport envelope.
$\hat{T}_{\text{tot}}^{\mu\nu}$	Notational total-EMT representative; inside the matching category it is required to be the attached operator $\hat{\mathbf{T}}_x^{\mu\nu}$.
$\mathbf{T}^{\mu\nu}$	Accepted Module A source tensor or profile distribution.

Symbol	Meaning
P_{profile}^{μ}	Integrated charge of a selected source profile.
$\hat{P}^{\mu}$	Self-adjoint Poincaré translation generator.
m_{Cas}	Positive stable-sector Poincaré Casimir mass.
p_{ω}^{μ}	Selected-state expectation four-momentum.
M_{src}	Energy mass of a localized stationary source profile.
$\Theta_x^{\mu\nu}(p', p)$	Complete total-EMT stable-sector matrix element.
$A_x(t)$	Scalar momentum/energy form factor; $A_x(0) = 1$.
$D_x(t)$	Stress/finite-size form factor.
m_{ff}	Intrinsic zero-transfer source mass; equals m_{Cas} .
$\mathcal{A}_g$	Intrinsic first-response active metric amplitude.
m_g	Active source mass coordinate after source calibration.
M_{B}	Bargmann central mass operator.
m_{i}	Inertial mass coordinate in the Bargmann/Schrödinger projection.
α_D	One projection-wide active/inertial proportionality factor.
$\mathcal{L}_D^P$	One-dimensional intrinsic Casimir/Noether charge line.
$\mathcal{L}_D^i$	Bargmann inertial readout line.
$\mathcal{L}_D^g$	Active zero-transfer metric-source readout line.
$\mathcal{S}(\mathbb{R}^3)$	Common invariant Schwartz core $\mathcal{S}(\mathbb{R}^3)$ for the explicit contraction.

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